

Dataset	Model	Metric	Normal	TransE+	DistMult+	Hybrid	Improvement
Llama2-conversation	BPR	Accuracy	0.7098	0.672	0.663	0.6603	-7.00%
Llama2-conversation	BPR	Precision@5	0.7826	0.838	0.8301	0.8241	+5.3%
Llama2-conversation	BPR	Precision@10	0.7591	0.8183	0.8162	0.8107	+6.8%
Llama2-conversation	BPR	Recall@5	0.4386	0.5629	0.5629	0.5625	+28.3%
Llama2-conversation	BPR	Recall@10	0.544	0.6232	0.6243	0.6228	+14.5%
Llama2-conversation	BPR	Ndcg@5	0.8322	0.8394	0.8379	0.8314	-0.10%
Llama2-conversation	BPR	Ndcg@10	0.8284	0.8285	0.829	0.8232	-0.60%
Llama2-conversation	BPR	Auc	0.5	0.7143	0.6947	0.6909	+38.2%
Llama2-conversation	BPR	Mrr	0.5	0.9122	0.9261	0.9191	+83.8%
Llama2-conversation	BPR	HIT@RATE@5	0.9938	0.9967	0.9983	0.995	+0.1%
Llama2-conversation	BPR	HIT@RATE@10	0.9987	1	1	0.9983	0.00%
Llama2-conversation	BPR	Map@5	0.7826	0.891	0.8977	0.8911	+13.9%
Llama2-conversation	BPR	Map@10	0.7591	0.8705	0.8712	0.8674	+14.3%
Llama2-conversation	COMPGCN	Accuracy	0.5069	0.6836	0.6889	0.6875	+35.6%
Llama2-conversation	COMPGCN	Precision@5	0.4887	0.8542	0.8588	0.8562	+75.2%
Llama2-conversation	COMPGCN	Precision@10	0.4899	0.8385	0.8393	0.8393	+71.3%
Llama2-conversation	COMPGCN	Recall@5	0.7148	0.5638	0.5642	0.5643	-21.10%
Llama2-conversation	COMPGCN	Recall@10	0.8816	0.6254	0.6258	0.6252	-29.10%
Llama2-conversation	COMPGCN	Ndcg@5	0.6996	0.8593	0.8623	0.8591	+22.8%
Llama2-conversation	COMPGCN	Ndcg@10	0.7845	0.8499	0.8508	0.8496	+8.3%
Llama2-conversation	COMPGCN	Auc	0.501	0.7405	0.7411	0.7409	+47.9%
Llama2-conversation	COMPGCN	Mrr	0.7302	0.935	0.9302	0.9311	+27.5%
Llama2-conversation	COMPGCN	HIT@RATE@5	0.8815	0.9967	0.9983	0.9983	+13.3%
Llama2-conversation	COMPGCN	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-conversation	COMPGCN	Map@5	0.6392	0.908	0.9095	0.9068	+41.9%
Llama2-conversation	COMPGCN	Map@10	0.6077	0.8872	0.8887	0.8862	+45.8%
Llama2-conversation	DEEPPFM	Accuracy	0.5055	0.6793	0.6766	0.6761	+33.8%
Llama2-conversation	DEEPPFM	Precision@5	0.4898	0.8562	0.8605	0.8555	+74.7%
Llama2-conversation	DEEPPFM	Precision@10	0.4899	0.837	0.8359	0.8365	+70.8%
Llama2-conversation	DEEPPFM	Recall@5	0.7158	0.5644	0.5645	0.564	-21.20%
Llama2-conversation	DEEPPFM	Recall@10	0.8815	0.625	0.6251	0.6257	-29.00%
Llama2-conversation	DEEPPFM	Ndcg@5	0.7007	0.8584	0.8636	0.8594	+22.7%
Llama2-conversation	DEEPPFM	Ndcg@10	0.7847	0.8478	0.8487	0.8483	+8.1%
Llama2-conversation	DEEPPFM	Auc	0.5047	0.7263	0.7202	0.7172	+42.1%
Llama2-conversation	DEEPPFM	Mrr	0.7297	0.931	0.9277	0.9328	+27.8%
Llama2-conversation	DEEPPFM	HIT@RATE@5	0.8828	1	1	1	+13.3%
Llama2-conversation	DEEPPFM	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-conversation	DEEPPFM	Map@5	0.643	0.9052	0.9124	0.9062	+40.9%
Llama2-conversation	DEEPPFM	Map@10	0.6075	0.8863	0.8893	0.8874	+46.1%
Llama2-conversation	GAT	Accuracy	0.5069	0.6874	0.6739	0.6821	+34.6%
Llama2-conversation	GAT	Precision@5	0.4881	0.8598	0.8486	0.8559	+75.3%
Llama2-conversation	GAT	Precision@10	0.4896	0.8375	0.8365	0.8347	+70.5%
Llama2-conversation	GAT	Recall@5	0.715	0.5647	0.5636	0.5641	-21.10%
Llama2-conversation	GAT	Recall@10	0.8811	0.6256	0.625	0.625	-29.10%
Llama2-conversation	GAT	Ndcg@5	0.698	0.864	0.8539	0.8599	+23.2%
Llama2-conversation	GAT	Ndcg@10	0.7827	0.8504	0.8471	0.8472	+8.2%
Llama2-conversation	GAT	Auc	0.4989	0.7384	0.724	0.7304	+46.4%
Llama2-conversation	GAT	Mrr	0.7279	0.9339	0.9302	0.93	+27.8%
Llama2-conversation	GAT	HIT@RATE@5	0.884	1	1	1	+13.1%
Llama2-conversation	GAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	GAT	Map@5	0.643	0.9108	0.9064	0.909	+41.4%
Llama2-conversation	GAT	Map@10	0.6052	0.8897	0.8839	0.8875	+46.6%
Llama2-conversation	GCN	Accuracy	0.5069	0.682	0.6755	0.6789	+33.9%
Llama2-conversation	GCN	Precision@5	0.4901	0.8559	0.8585	0.8598	+75.4%
Llama2-conversation	GCN	Precision@10	0.4899	0.8393	0.8413	0.8354	+70.5%
Llama2-conversation	GCN	Recall@5	0.7181	0.5634	0.5642	0.5643	-21.40%
Llama2-conversation	GCN	Recall@10	0.8817	0.6255	0.6258	0.6248	-29.10%
Llama2-conversation	GCN	Ndcg@5	0.7019	0.861	0.8633	0.8609	+22.7%
Llama2-conversation	GCN	Ndcg@10	0.7852	0.8512	0.8531	0.8468	+7.9%

Llama2-conversation	GCN	Auc	0.5027	0.732	0.7229	0.7262	+44.5%
Llama2-conversation	GCN	Mrr	0.7307	0.9319	0.9343	0.9249	+26.6%
Llama2-conversation	GCN	HIT@RATE@5	0.8855	1	1	1	+12.9%
Llama2-conversation	GCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	GCN	Map@5	0.6434	0.9111	0.9123	0.9069	+40.9%
Llama2-conversation	GCN	Map@10	0.6171	0.8894	0.8903	0.8872	+43.8%
Llama2-conversation	GRAPHSAGE	Accuracy	0.6908	0.6838	0.6726	0.6786	-1.80%
Llama2-conversation	GRAPHSAGE	Precision@5	0.7639	0.8532	0.8526	0.8562	+12.1%
Llama2-conversation	GRAPHSAGE	Precision@10	0.7393	0.838	0.8387	0.837	+13.2%
Llama2-conversation	GRAPHSAGE	Recall@5	0.4333	0.5641	0.5638	0.564	+30.2%
Llama2-conversation	GRAPHSAGE	Recall@10	0.5326	0.625	0.6252	0.6256	+17.5%
Llama2-conversation	GRAPHSAGE	Ndcg@5	0.813	0.8565	0.8537	0.8588	+5.6%
Llama2-conversation	GRAPHSAGE	Ndcg@10	0.8052	0.8483	0.847	0.8483	+5.4%
Llama2-conversation	GRAPHSAGE	Auc	0.5	0.7321	0.7151	0.7279	+45.6%
Llama2-conversation	GRAPHSAGE	Mrr	0.5	0.9267	0.9206	0.9258	+85.2%
Llama2-conversation	GRAPHSAGE	HIT@RATE@5	0.9931	1	0.9983	1	+0.7%
Llama2-conversation	GRAPHSAGE	HIT@RATE@10	0.9976	1	1	1	+0.2%
Llama2-conversation	GRAPHSAGE	Map@5	0.7639	0.9065	0.9019	0.9068	+18.7%
Llama2-conversation	GRAPHSAGE	Map@10	0.7393	0.8867	0.8844	0.8885	+20.2%
Llama2-conversation	LIGHTGCN	Accuracy	0.5076	0.6803	0.6824	0.6822	+34.4%
Llama2-conversation	LIGHTGCN	Precision@5	0.491	0.8483	0.8595	0.8479	+72.7%
Llama2-conversation	LIGHTGCN	Precision@10	0.4896	0.8342	0.84	0.8331	+70.2%
Llama2-conversation	LIGHTGCN	Recall@5	0.7187	0.5638	0.5643	0.5636	-21.60%
Llama2-conversation	LIGHTGCN	Recall@10	0.8809	0.6252	0.6254	0.6248	-29.10%
Llama2-conversation	LIGHTGCN	Ndcg@5	0.7051	0.8526	0.863	0.8526	+20.9%
Llama2-conversation	LIGHTGCN	Ndcg@10	0.7871	0.8451	0.8518	0.8441	+7.2%
Llama2-conversation	LIGHTGCN	Auc	0.5013	0.7265	0.7232	0.7317	+46.0%
Llama2-conversation	LIGHTGCN	Mrr	0.7348	0.9288	0.9277	0.9318	+26.8%
Llama2-conversation	LIGHTGCN	HIT@RATE@5	0.8855	1	1	1	+12.9%
Llama2-conversation	LIGHTGCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	LIGHTGCN	Map@5	0.647	0.9047	0.9126	0.9042	+39.7%
Llama2-conversation	LIGHTGCN	Map@10	0.6108	0.8833	0.891	0.8825	+44.5%
Llama2-conversation	KGAT	Accuracy	0.5069	0.6893	0.6833	0.6839	+34.9%
Llama2-conversation	KGAT	Precision@5	0.4881	0.8615	0.8509	0.8559	+75.3%
Llama2-conversation	KGAT	Precision@10	0.4896	0.837	0.836	0.835	+70.6%
Llama2-conversation	KGAT	Recall@5	0.715	0.5647	0.5636	0.5644	-21.10%
Llama2-conversation	KGAT	Recall@10	0.8811	0.6252	0.6248	0.6254	-29.00%
Llama2-conversation	KGAT	Ndcg@5	0.698	0.8626	0.8563	0.8568	+22.8%
Llama2-conversation	KGAT	Ndcg@10	0.7827	0.8486	0.8476	0.8456	+8.0%
Llama2-conversation	KGAT	Auc	0.4989	0.7394	0.7354	0.7346	+47.2%
Llama2-conversation	KGAT	Mrr	0.7279	0.9269	0.9316	0.9256	+27.2%
Llama2-conversation	KGAT	HIT@RATE@5	0.884	0.9983	0.9967	0.9983	+12.9%
Llama2-conversation	KGAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	KGAT	Map@5	0.643	0.9066	0.9071	0.9034	+40.5%
Llama2-conversation	KGAT	Map@10	0.6052	0.8881	0.8862	0.8851	+46.2%
Llama2-conversation	KGIN	Accuracy	0.5069	0.689	0.6818	0.684	+34.9%
Llama2-conversation	KGIN	Precision@5	0.4906	0.8592	0.8579	0.8605	+75.4%
Llama2-conversation	KGIN	Precision@10	0.4899	0.8329	0.8319	0.8377	+71.0%
Llama2-conversation	KGIN	Recall@5	0.7181	0.5647	0.5648	0.5648	-21.30%
Llama2-conversation	KGIN	Recall@10	0.8816	0.6247	0.6246	0.6254	-29.10%
Llama2-conversation	KGIN	Ndcg@5	0.7018	0.862	0.8613	0.8647	+23.2%
Llama2-conversation	KGIN	Ndcg@10	0.7848	0.8462	0.8456	0.8507	+8.4%
Llama2-conversation	KGIN	Auc	0.5054	0.7385	0.7282	0.7338	+45.2%
Llama2-conversation	KGIN	Mrr	0.7305	0.9306	0.9297	0.9342	+27.9%
Llama2-conversation	KGIN	HIT@RATE@5	0.8847	1	1	1	+13.0%
Llama2-conversation	KGIN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	KGIN	Map@5	0.6462	0.9086	0.9106	0.9135	+41.4%
Llama2-conversation	KGIN	Map@10	0.6161	0.8877	0.8881	0.8902	+44.5%
Llama2-conversation	MULTIGCCF	Accuracy	0.5066	0.6886	0.686	0.6812	+34.5%

Llama2-conversation	MULTIGCCF	Precision@5	0.4879	0.8555	0.8572	0.8585	+76.0%
Llama2-conversation	MULTIGCCF	Precision@10	0.4897	0.835	0.8426	0.8382	+71.2%
Llama2-conversation	MULTIGCCF	Recall@5	0.7139	0.564	0.5643	0.5642	-21.00%
Llama2-conversation	MULTIGCCF	Recall@10	0.8811	0.6249	0.6263	0.6256	-29.00%
Llama2-conversation	MULTIGCCF	Ndcg@5	0.6979	0.8597	0.8599	0.8609	+23.4%
Llama2-conversation	MULTIGCCF	Ndcg@10	0.7831	0.8475	0.8522	0.8495	+8.5%
Llama2-conversation	MULTIGCCF	Auc	0.5005	0.7395	0.7349	0.7308	+46.0%
Llama2-conversation	MULTIGCCF	Mrr	0.7293	0.9317	0.93	0.9312	+27.7%
Llama2-conversation	MULTIGCCF	HIT@RATE@5	0.8823	1	0.9983	1	+13.3%
Llama2-conversation	MULTIGCCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-conversation	MULTIGCCF	Map@5	0.636	0.9081	0.9073	0.9067	+42.6%
Llama2-conversation	MULTIGCCF	Map@10	0.6064	0.8869	0.8876	0.8881	+46.5%
Llama2-conversation	NEUMF	Accuracy	0.5069	0.6763	0.676	0.6783	+33.8%
Llama2-conversation	NEUMF	Precision@5	0.4917	0.8483	0.8483	0.8572	+74.3%
Llama2-conversation	NEUMF	Precision@10	0.4901	0.8331	0.8253	0.8344	+70.2%
Llama2-conversation	NEUMF	Recall@5	0.7193	0.5633	0.5631	0.5641	-21.60%
Llama2-conversation	NEUMF	Recall@10	0.8818	0.6243	0.6235	0.6249	-29.10%
Llama2-conversation	NEUMF	Ndcg@5	0.7068	0.8535	0.8546	0.8597	+21.6%
Llama2-conversation	NEUMF	Ndcg@10	0.7892	0.8444	0.8397	0.8464	+7.2%
Llama2-conversation	NEUMF	Auc	0.5053	0.7228	0.7141	0.7264	+43.8%
Llama2-conversation	NEUMF	Mrr	0.7369	0.9296	0.9327	0.9267	+25.8%
Llama2-conversation	NEUMF	HIT@RATE@5	0.8826	0.9983	0.9967	0.9983	+13.1%
Llama2-conversation	NEUMF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-conversation	NEUMF	Map@5	0.6509	0.9061	0.9067	0.9072	+39.4%
Llama2-conversation	NEUMF	Map@10	0.6133	0.8835	0.8832	0.8872	+44.7%
Llama2-conversation	NGCF	Accuracy	0.5066	0.6888	0.685	0.6848	+35.2%
Llama2-conversation	NGCF	Precision@5	0.4879	0.8545	0.8572	0.8621	+76.7%
Llama2-conversation	NGCF	Precision@10	0.4897	0.837	0.838	0.8372	+71.0%
Llama2-conversation	NGCF	Recall@5	0.7139	0.564	0.5643	0.5646	-20.90%
Llama2-conversation	NGCF	Recall@10	0.8811	0.6253	0.6255	0.6251	-29.00%
Llama2-conversation	NGCF	Ndcg@5	0.6979	0.8596	0.86	0.863	+23.7%
Llama2-conversation	NGCF	Ndcg@10	0.7831	0.8492	0.8491	0.8484	+8.3%
Llama2-conversation	NGCF	Auc	0.5005	0.7398	0.7328	0.734	+46.6%
Llama2-conversation	NGCF	Mrr	0.7293	0.9346	0.9317	0.9309	+27.6%
Llama2-conversation	NGCF	HIT@RATE@5	0.8823	1	0.9983	0.9983	+13.2%
Llama2-conversation	NGCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-conversation	NGCF	Map@5	0.636	0.9103	0.9081	0.9064	+42.5%
Llama2-conversation	NGCF	Map@10	0.6064	0.8873	0.8869	0.8868	+46.2%
Llama2-conversation	PINSAGE	Accuracy	0.5069	0.6899	0.6861	0.6833	+34.8%
Llama2-conversation	PINSAGE	Precision@5	0.4906	0.8579	0.8542	0.8565	+74.6%
Llama2-conversation	PINSAGE	Precision@10	0.4899	0.838	0.8364	0.8336	+70.1%
Llama2-conversation	PINSAGE	Recall@5	0.7181	0.5642	0.5638	0.564	-21.50%
Llama2-conversation	PINSAGE	Recall@10	0.8816	0.6253	0.6257	0.6251	-29.10%
Llama2-conversation	PINSAGE	Ndcg@5	0.7018	0.8607	0.8572	0.8599	+22.5%
Llama2-conversation	PINSAGE	Ndcg@10	0.7848	0.8494	0.8474	0.8463	+7.8%
Llama2-conversation	PINSAGE	Auc	0.5054	0.7391	0.7305	0.73	+44.4%
Llama2-conversation	PINSAGE	Mrr	0.7305	0.9279	0.9281	0.9303	+27.4%
Llama2-conversation	PINSAGE	HIT@RATE@5	0.8847	0.9983	0.9983	0.9983	+12.8%
Llama2-conversation	PINSAGE	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	PINSAGE	Map@5	0.6462	0.9088	0.9044	0.9068	+40.3%
Llama2-conversation	PINSAGE	Map@10	0.6161	0.888	0.8855	0.887	+44.0%
Llama2-conversation	SASREC	Accuracy	0.5069	0.6801	0.6768	0.6744	+33.0%
Llama2-conversation	SASREC	Precision@5	0.4907	0.8516	0.8562	0.8542	+74.1%
Llama2-conversation	SASREC	Precision@10	0.4895	0.8364	0.8413	0.8423	+72.1%
Llama2-conversation	SASREC	Recall@5	0.7185	0.5637	0.5638	0.5642	-21.50%
Llama2-conversation	SASREC	Recall@10	0.8809	0.6254	0.6256	0.626	-28.90%
Llama2-conversation	SASREC	Ndcg@5	0.7048	0.8572	0.8567	0.8567	+21.5%
Llama2-conversation	SASREC	Ndcg@10	0.7872	0.8481	0.8495	0.8508	+8.1%
Llama2-conversation	SASREC	Auc	0.4981	0.7297	0.7138	0.7164	+43.8%

Llama2-conversation	SASREC	Mrr	0.7351	0.9346	0.9228	0.9264	+26.0%
Llama2-conversation	SASREC	HIT@RATE@5	0.887	1	1	1	+12.7%
Llama2-conversation	SASREC	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	SASREC	Map@5	0.6446	0.9074	0.9014	0.9042	+40.3%
Llama2-conversation	SASREC	Map@10	0.6096	0.8861	0.885	0.8859	+45.3%
Llama2-conversation	SRGNN	Accuracy	0.5081	0.6876	0.6815	0.6788	+33.6%
Llama2-conversation	SRGNN	Precision@5	0.4896	0.8592	0.8572	0.8582	+75.3%
Llama2-conversation	SRGNN	Precision@10	0.4898	0.8403	0.8375	0.8374	+71.0%
Llama2-conversation	SRGNN	Recall@5	0.716	0.5645	0.5639	0.5643	-21.20%
Llama2-conversation	SRGNN	Recall@10	0.8815	0.6258	0.6243	0.6252	-29.10%
Llama2-conversation	SRGNN	Ndcg@5	0.7006	0.8603	0.8601	0.8624	+23.1%
Llama2-conversation	SRGNN	Ndcg@10	0.7845	0.85	0.8486	0.85	+8.3%
Llama2-conversation	SRGNN	Auc	0.5021	0.7388	0.7303	0.729	+45.2%
Llama2-conversation	SRGNN	Mrr	0.7334	0.9247	0.9292	0.9346	+27.4%
Llama2-conversation	SRGNN	HIT@RATE@5	0.8838	0.9983	0.9983	1	+13.1%
Llama2-conversation	SRGNN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-conversation	SRGNN	Map@5	0.6472	0.906	0.9068	0.9103	+40.7%
Llama2-conversation	SRGNN	Map@10	0.6192	0.8879	0.887	0.8899	+43.7%
Llama2-structured	BPR	Accuracy	0.7098	0.6722	0.6518	0.6574	-7.40%
Llama2-structured	BPR	Precision@5	0.7826	0.8367	0.838	0.8344	+6.6%
Llama2-structured	BPR	Precision@10	0.7591	0.8221	0.8179	0.8117	+6.9%
Llama2-structured	BPR	Recall@5	0.4386	0.5632	0.5634	0.5634	+28.5%
Llama2-structured	BPR	Recall@10	0.544	0.6236	0.6241	0.6233	+14.6%
Llama2-structured	BPR	Ndcg@5	0.8322	0.8407	0.8395	0.8396	+0.9%
Llama2-structured	BPR	Ndcg@10	0.8284	0.8323	0.8282	0.8254	-0.40%
Llama2-structured	BPR	Auc	0.5	0.7132	0.686	0.6885	+37.7%
Llama2-structured	BPR	Mrr	0.5	0.9203	0.9148	0.9198	+84.0%
Llama2-structured	BPR	HIT@RATE@5	0.9938	0.9983	0.9967	0.9967	+0.3%
Llama2-structured	BPR	HIT@RATE@10	0.9987	1	0.9967	1	+0.1%
Llama2-structured	BPR	Map@5	0.7826	0.8946	0.891	0.897	+14.6%
Llama2-structured	BPR	Map@10	0.7591	0.8717	0.87	0.8714	+14.8%
Llama2-structured	COMPGCN	Accuracy	0.5069	0.6781	0.6845	0.6848	+35.1%
Llama2-structured	COMPGCN	Precision@5	0.4887	0.8575	0.8555	0.8552	+75.0%
Llama2-structured	COMPGCN	Precision@10	0.4899	0.8387	0.8412	0.84	+71.5%
Llama2-structured	COMPGCN	Recall@5	0.7148	0.5641	0.5644	0.5644	-21.00%
Llama2-structured	COMPGCN	Recall@10	0.8816	0.6257	0.6261	0.6255	-29.00%
Llama2-structured	COMPGCN	Ndcg@5	0.6996	0.8589	0.8598	0.8605	+23.0%
Llama2-structured	COMPGCN	Ndcg@10	0.7845	0.8485	0.8517	0.8515	+8.5%
Llama2-structured	COMPGCN	Auc	0.501	0.7398	0.735	0.7348	+46.7%
Llama2-structured	COMPGCN	Mrr	0.7302	0.929	0.9331	0.9339	+27.9%
Llama2-structured	COMPGCN	HIT@RATE@5	0.8815	1	0.9983	0.9983	+13.3%
Llama2-structured	COMPGCN	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-structured	COMPGCN	Map@5	0.6392	0.9051	0.9078	0.9092	+42.2%
Llama2-structured	COMPGCN	Map@10	0.6077	0.8862	0.8876	0.888	+46.1%
Llama2-structured	DEEPPM	Accuracy	0.5055	0.68	0.6736	0.6783	+34.2%
Llama2-structured	DEEPPM	Precision@5	0.4898	0.8506	0.8579	0.8545	+74.5%
Llama2-structured	DEEPPM	Precision@10	0.4899	0.8359	0.8307	0.8405	+71.6%
Llama2-structured	DEEPPM	Recall@5	0.7158	0.5636	0.5644	0.5639	-21.20%
Llama2-structured	DEEPPM	Recall@10	0.8815	0.6253	0.6234	0.6254	-29.00%
Llama2-structured	DEEPPM	Ndcg@5	0.7007	0.8543	0.8617	0.8559	+22.1%
Llama2-structured	DEEPPM	Ndcg@10	0.7847	0.8463	0.8449	0.8492	+8.2%
Llama2-structured	DEEPPM	Auc	0.5047	0.7258	0.7187	0.7213	+42.9%
Llama2-structured	DEEPPM	Mrr	0.7297	0.9256	0.9314	0.9222	+26.4%
Llama2-structured	DEEPPM	HIT@RATE@5	0.8828	0.9983	1	1	+13.3%
Llama2-structured	DEEPPM	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-structured	DEEPPM	Map@5	0.643	0.9045	0.9101	0.9038	+40.6%
Llama2-structured	DEEPPM	Map@10	0.6075	0.8832	0.8871	0.8853	+45.7%
Llama2-structured	GAT	Accuracy	0.5069	0.6882	0.6846	0.6808	+34.3%
Llama2-structured	GAT	Precision@5	0.4881	0.8562	0.8605	0.8602	+76.2%

Llama2-structured	GAT	Precision@10	0.4896	0.8388	0.8395	0.8398	+71.5%
Llama2-structured	GAT	Recall@5	0.715	0.5645	0.5642	0.5647	-21.00%
Llama2-structured	GAT	Recall@10	0.8811	0.6257	0.6261	0.6259	-29.00%
Llama2-structured	GAT	Ndcg@5	0.698	0.8581	0.8627	0.8614	+23.4%
Llama2-structured	GAT	Ndcg@10	0.7827	0.8488	0.8505	0.8501	+8.6%
Llama2-structured	GAT	Auc	0.4989	0.7378	0.735	0.7302	+46.4%
Llama2-structured	GAT	Mrr	0.7279	0.9265	0.9293	0.9282	+27.5%
Llama2-structured	GAT	HIT@RATE@5	0.884	1	1	1	+13.1%
Llama2-structured	GAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	GAT	Map@5	0.643	0.9048	0.9086	0.9063	+41.0%
Llama2-structured	GAT	Map@10	0.6052	0.8853	0.888	0.8872	+46.6%
Llama2-structured	GCN	Accuracy	0.5069	0.6823	0.6809	0.6791	+34.0%
Llama2-structured	GCN	Precision@5	0.4901	0.8552	0.8549	0.8555	+74.6%
Llama2-structured	GCN	Precision@10	0.4899	0.8377	0.8367	0.8393	+71.3%
Llama2-structured	GCN	Recall@5	0.7181	0.5639	0.5642	0.5638	-21.50%
Llama2-structured	GCN	Recall@10	0.8817	0.6252	0.6253	0.6259	-29.00%
Llama2-structured	GCN	Ndcg@5	0.7019	0.8591	0.8598	0.8592	+22.4%
Llama2-structured	GCN	Ndcg@10	0.7852	0.849	0.8486	0.8498	+8.2%
Llama2-structured	GCN	Auc	0.5027	0.7267	0.7279	0.7202	+43.3%
Llama2-structured	GCN	Mrr	0.7307	0.9321	0.9293	0.9316	+27.5%
Llama2-structured	GCN	HIT@RATE@5	0.8855	1	1	1	+12.9%
Llama2-structured	GCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	GCN	Map@5	0.6434	0.9079	0.9105	0.909	+41.3%
Llama2-structured	GCN	Map@10	0.6171	0.8873	0.8873	0.8865	+43.7%
Llama2-structured	GRAPHSAGE	Accuracy	0.6908	0.6851	0.6764	0.6732	-2.50%
Llama2-structured	GRAPHSAGE	Precision@5	0.7639	0.8618	0.8539	0.8579	+12.3%
Llama2-structured	GRAPHSAGE	Precision@10	0.7393	0.8354	0.8382	0.837	+13.2%
Llama2-structured	GRAPHSAGE	Recall@5	0.4333	0.5644	0.5642	0.5639	+30.1%
Llama2-structured	GRAPHSAGE	Recall@10	0.5326	0.6246	0.6254	0.6247	+17.3%
Llama2-structured	GRAPHSAGE	Ndcg@5	0.813	0.8653	0.8586	0.8624	+6.1%
Llama2-structured	GRAPHSAGE	Ndcg@10	0.8052	0.8488	0.8495	0.8497	+5.5%
Llama2-structured	GRAPHSAGE	Auc	0.5	0.7343	0.7249	0.7158	+43.2%
Llama2-structured	GRAPHSAGE	Mrr	0.5	0.9356	0.932	0.9371	+87.4%
Llama2-structured	GRAPHSAGE	HIT@RATE@5	0.9931	0.9983	0.9967	0.9983	+0.5%
Llama2-structured	GRAPHSAGE	HIT@RATE@10	0.9976	1	1	1	+0.2%
Llama2-structured	GRAPHSAGE	Map@5	0.7639	0.9114	0.9074	0.9121	+19.4%
Llama2-structured	GRAPHSAGE	Map@10	0.7393	0.8892	0.8868	0.8892	+20.3%
Llama2-structured	LIGHTGCN	Accuracy	0.5076	0.6859	0.6747	0.6782	+33.6%
Llama2-structured	LIGHTGCN	Precision@5	0.491	0.8542	0.8552	0.8499	+73.1%
Llama2-structured	LIGHTGCN	Precision@10	0.4896	0.8382	0.8385	0.8302	+69.6%
Llama2-structured	LIGHTGCN	Recall@5	0.7187	0.5637	0.564	0.5633	-21.60%
Llama2-structured	LIGHTGCN	Recall@10	0.8809	0.6252	0.6256	0.6245	-29.10%
Llama2-structured	LIGHTGCN	Ndcg@5	0.7051	0.8587	0.8586	0.8526	+20.9%
Llama2-structured	LIGHTGCN	Ndcg@10	0.7871	0.8493	0.8493	0.8416	+6.9%
Llama2-structured	LIGHTGCN	Auc	0.5013	0.7332	0.7195	0.7178	+43.2%
Llama2-structured	LIGHTGCN	Mrr	0.7348	0.9336	0.9329	0.9243	+25.8%
Llama2-structured	LIGHTGCN	HIT@RATE@5	0.8855	1	1	0.9983	+12.7%
Llama2-structured	LIGHTGCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	LIGHTGCN	Map@5	0.647	0.9086	0.9074	0.9037	+39.7%
Llama2-structured	LIGHTGCN	Map@10	0.6108	0.8866	0.8864	0.8821	+44.4%
Llama2-structured	KGAT	Accuracy	0.5069	0.6881	0.6866	0.6786	+33.9%
Llama2-structured	KGAT	Precision@5	0.4881	0.8539	0.8539	0.8555	+75.3%
Llama2-structured	KGAT	Precision@10	0.4896	0.8379	0.8393	0.8377	+71.1%
Llama2-structured	KGAT	Recall@5	0.715	0.564	0.5636	0.5636	-21.20%
Llama2-structured	KGAT	Recall@10	0.8811	0.6248	0.6261	0.6253	-29.00%
Llama2-structured	KGAT	Ndcg@5	0.698	0.8568	0.8586	0.8607	+23.3%
Llama2-structured	KGAT	Ndcg@10	0.7827	0.8479	0.8504	0.8499	+8.6%
Llama2-structured	KGAT	Auc	0.4989	0.7378	0.739	0.7236	+45.0%
Llama2-structured	KGAT	Mrr	0.7279	0.9273	0.9348	0.9341	+28.3%

Llama2-structured	KGAT	HIT@RATE@5	0.884	0.9983	1	1	+13.1%
Llama2-structured	KGAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	KGAT	Map@5	0.643	0.905	0.9073	0.9109	+41.7%
Llama2-structured	KGAT	Map@10	0.6052	0.8854	0.8871	0.8876	+46.7%
Llama2-structured	KGIN	Accuracy	0.5069	0.6891	0.6835	0.6856	+35.3%
Llama2-structured	KGIN	Precision@5	0.4906	0.8575	0.8552	0.8559	+74.5%
Llama2-structured	KGIN	Precision@10	0.4899	0.8375	0.8385	0.838	+71.1%
Llama2-structured	KGIN	Recall@5	0.7181	0.5639	0.5639	0.5646	-21.40%
Llama2-structured	KGIN	Recall@10	0.8816	0.6257	0.6259	0.6258	-29.00%
Llama2-structured	KGIN	Ndcg@5	0.7018	0.8605	0.8578	0.859	+22.4%
Llama2-structured	KGIN	Ndcg@10	0.7848	0.8489	0.8488	0.849	+8.2%
Llama2-structured	KGIN	Auc	0.5054	0.7393	0.7284	0.7348	+45.4%
Llama2-structured	KGIN	Mrr	0.7305	0.9306	0.9258	0.9308	+27.4%
Llama2-structured	KGIN	HIT@RATE@5	0.8847	1	1	1	+13.0%
Llama2-structured	KGIN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	KGIN	Map@5	0.6462	0.9081	0.9062	0.9074	+40.4%
Llama2-structured	KGIN	Map@10	0.6161	0.8879	0.8862	0.8863	+43.9%
Llama2-structured	MULTIGCCF	Accuracy	0.5066	0.6878	0.6869	0.6843	+35.1%
Llama2-structured	MULTIGCCF	Precision@5	0.4879	0.8555	0.8595	0.8575	+75.8%
Llama2-structured	MULTIGCCF	Precision@10	0.4897	0.8369	0.8385	0.8354	+70.6%
Llama2-structured	MULTIGCCF	Recall@5	0.7139	0.5638	0.564	0.5642	-21.00%
Llama2-structured	MULTIGCCF	Recall@10	0.8811	0.6252	0.6255	0.6252	-29.00%
Llama2-structured	MULTIGCCF	Ndcg@5	0.6979	0.8589	0.8623	0.8597	+23.2%
Llama2-structured	MULTIGCCF	Ndcg@10	0.7831	0.8482	0.8501	0.847	+8.2%
Llama2-structured	MULTIGCCF	Auc	0.5005	0.7393	0.7349	0.7325	+46.4%
Llama2-structured	MULTIGCCF	Mrr	0.7293	0.9288	0.9319	0.9258	+26.9%
Llama2-structured	MULTIGCCF	HIT@RATE@5	0.8823	0.9983	1	1	+13.3%
Llama2-structured	MULTIGCCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-structured	MULTIGCCF	Map@5	0.636	0.9067	0.9085	0.9067	+42.6%
Llama2-structured	MULTIGCCF	Map@10	0.6064	0.8868	0.8887	0.8875	+46.4%
Llama2-structured	NEUMF	Accuracy	0.5069	0.6833	0.6794	0.6813	+34.4%
Llama2-structured	NEUMF	Precision@5	0.4917	0.8562	0.8526	0.8509	+73.1%
Llama2-structured	NEUMF	Precision@10	0.4901	0.8382	0.8344	0.8334	+70.0%
Llama2-structured	NEUMF	Recall@5	0.7193	0.5642	0.5637	0.5636	-21.60%
Llama2-structured	NEUMF	Recall@10	0.8818	0.6255	0.6248	0.625	-29.10%
Llama2-structured	NEUMF	Ndcg@5	0.7068	0.8591	0.8575	0.8554	+21.0%
Llama2-structured	NEUMF	Ndcg@10	0.7892	0.8488	0.8467	0.8456	+7.1%
Llama2-structured	NEUMF	Auc	0.5053	0.7333	0.7272	0.7233	+43.1%
Llama2-structured	NEUMF	Mrr	0.7369	0.9249	0.9313	0.9302	+26.2%
Llama2-structured	NEUMF	HIT@RATE@5	0.8826	0.9967	0.9983	1	+13.3%
Llama2-structured	NEUMF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-structured	NEUMF	Map@5	0.6509	0.9082	0.9085	0.9067	+39.3%
Llama2-structured	NEUMF	Map@10	0.6133	0.8865	0.8876	0.8861	+44.5%
Llama2-structured	NGCF	Accuracy	0.5066	0.6881	0.6859	0.6892	+36.0%
Llama2-structured	NGCF	Precision@5	0.4879	0.8572	0.8549	0.8592	+76.1%
Llama2-structured	NGCF	Precision@10	0.4897	0.8387	0.8398	0.8364	+70.8%
Llama2-structured	NGCF	Recall@5	0.7139	0.5639	0.5641	0.5643	-21.00%
Llama2-structured	NGCF	Recall@10	0.8811	0.6253	0.6255	0.625	-29.10%
Llama2-structured	NGCF	Ndcg@5	0.6979	0.8614	0.8592	0.8612	+23.4%
Llama2-structured	NGCF	Ndcg@10	0.7831	0.8504	0.8505	0.8479	+8.3%
Llama2-structured	NGCF	Auc	0.5005	0.7386	0.7368	0.7381	+47.5%
Llama2-structured	NGCF	Mrr	0.7293	0.9304	0.9316	0.9288	+27.4%
Llama2-structured	NGCF	HIT@RATE@5	0.8823	0.9983	1	1	+13.3%
Llama2-structured	NGCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Llama2-structured	NGCF	Map@5	0.636	0.91	0.9083	0.9071	+42.6%
Llama2-structured	NGCF	Map@10	0.6064	0.8886	0.8866	0.8872	+46.3%
Llama2-structured	PINSAGE	Accuracy	0.5069	0.6872	0.685	0.6834	+34.8%
Llama2-structured	PINSAGE	Precision@5	0.4906	0.8562	0.8559	0.8545	+74.2%
Llama2-structured	PINSAGE	Precision@10	0.4899	0.839	0.8339	0.8402	+71.5%

Llama2-structured	PINSAGE	Recall@5	0.7181	0.564	0.5645	0.5639	-21.50%
Llama2-structured	PINSAGE	Recall@10	0.8816	0.6256	0.6251	0.6257	-29.00%
Llama2-structured	PINSAGE	Ndcg@5	0.7018	0.8594	0.8605	0.8584	+22.3%
Llama2-structured	PINSAGE	Ndcg@10	0.7848	0.8497	0.847	0.8504	+8.4%
Llama2-structured	PINSAGE	Auc	0.5054	0.7401	0.7304	0.7339	+45.2%
Llama2-structured	PINSAGE	Mrr	0.7305	0.9281	0.9342	0.9304	+27.4%
Llama2-structured	PINSAGE	HIT@RATE@5	0.8847	1	0.9983	1	+13.0%
Llama2-structured	PINSAGE	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	PINSAGE	Map@5	0.6462	0.9083	0.9084	0.907	+40.4%
Llama2-structured	PINSAGE	Map@10	0.6161	0.8872	0.886	0.8868	+43.9%
Llama2-structured	SASREC	Accuracy	0.5069	0.6871	0.6812	0.6768	+33.5%
Llama2-structured	SASREC	Precision@5	0.4907	0.8539	0.8493	0.8552	+74.3%
Llama2-structured	SASREC	Precision@10	0.4895	0.8403	0.8388	0.8425	+72.1%
Llama2-structured	SASREC	Recall@5	0.7185	0.5639	0.5634	0.5642	-21.50%
Llama2-structured	SASREC	Recall@10	0.8809	0.6254	0.6255	0.6262	-28.90%
Llama2-structured	SASREC	Ndcg@5	0.7048	0.8566	0.8555	0.8566	+21.5%
Llama2-structured	SASREC	Ndcg@10	0.7872	0.8496	0.8494	0.8507	+8.1%
Llama2-structured	SASREC	Auc	0.4981	0.7313	0.7254	0.7274	+46.0%
Llama2-structured	SASREC	Mrr	0.7351	0.925	0.9308	0.924	+25.7%
Llama2-structured	SASREC	HIT@RATE@5	0.887	1	1	1	+12.7%
Llama2-structured	SASREC	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	SASREC	Map@5	0.6446	0.905	0.9078	0.9026	+40.0%
Llama2-structured	SASREC	Map@10	0.6096	0.8865	0.8856	0.8862	+45.4%
Llama2-structured	SRGNN	Accuracy	0.5081	0.6852	0.6805	0.6817	+34.2%
Llama2-structured	SRGNN	Precision@5	0.4896	0.8575	0.8562	0.8585	+75.3%
Llama2-structured	SRGNN	Precision@10	0.4898	0.8393	0.841	0.8364	+70.8%
Llama2-structured	SRGNN	Recall@5	0.716	0.5644	0.564	0.5641	-21.20%
Llama2-structured	SRGNN	Recall@10	0.8815	0.6257	0.6263	0.6254	-29.10%
Llama2-structured	SRGNN	Ndcg@5	0.7006	0.862	0.8596	0.8602	+22.8%
Llama2-structured	SRGNN	Ndcg@10	0.7845	0.8513	0.8515	0.8477	+8.1%
Llama2-structured	SRGNN	Auc	0.5021	0.7356	0.7258	0.7283	+45.0%
Llama2-structured	SRGNN	Mrr	0.7334	0.9291	0.929	0.9322	+27.1%
Llama2-structured	SRGNN	HIT@RATE@5	0.8838	1	0.9983	1	+13.1%
Llama2-structured	SRGNN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Llama2-structured	SRGNN	Map@5	0.6472	0.9126	0.9081	0.9062	+40.0%
Llama2-structured	SRGNN	Map@10	0.6192	0.8903	0.8885	0.8866	+43.2%
Gemma3-conversation	BPR	Accuracy	0.7098	0.6804	0.6532	0.6497	-8.50%
Gemma3-conversation	BPR	Precision@5	0.7826	0.8483	0.8407	0.8307	+6.2%
Gemma3-conversation	BPR	Precision@10	0.7591	0.8322	0.8154	0.8167	+7.6%
Gemma3-conversation	BPR	Recall@5	0.4386	0.5634	0.5638	0.5627	+28.3%
Gemma3-conversation	BPR	Recall@10	0.544	0.6247	0.6243	0.6239	+14.7%
Gemma3-conversation	BPR	Ndcg@5	0.8322	0.8535	0.8404	0.8341	+0.2%
Gemma3-conversation	BPR	Ndcg@10	0.8284	0.8441	0.8263	0.8269	-0.20%
Gemma3-conversation	BPR	Auc	0.5	0.7251	0.6775	0.6721	+34.4%
Gemma3-conversation	BPR	Mrr	0.5	0.9279	0.9134	0.9155	+83.1%
Gemma3-conversation	BPR	HIT@RATE@5	0.9938	1	0.9967	0.9983	+0.5%
Gemma3-conversation	BPR	HIT@RATE@10	0.9987	1	1	1	+0.1%
Gemma3-conversation	BPR	Map@5	0.7826	0.9053	0.8885	0.8908	+13.8%
Gemma3-conversation	BPR	Map@10	0.7591	0.8835	0.8707	0.869	+14.5%
Gemma3-conversation	COMPGCN	Accuracy	0.5069	0.6785	0.6923	0.6845	+35.0%
Gemma3-conversation	COMPGCN	Precision@5	0.4887	0.8569	0.8579	0.8519	+74.3%
Gemma3-conversation	COMPGCN	Precision@10	0.4899	0.8369	0.8369	0.8402	+71.5%
Gemma3-conversation	COMPGCN	Recall@5	0.7148	0.5638	0.5641	0.5638	-21.10%
Gemma3-conversation	COMPGCN	Recall@10	0.8816	0.6251	0.625	0.6257	-29.00%
Gemma3-conversation	COMPGCN	Ndcg@5	0.6996	0.859	0.8596	0.8559	+22.3%
Gemma3-conversation	COMPGCN	Ndcg@10	0.7845	0.8476	0.8477	0.8498	+8.3%
Gemma3-conversation	COMPGCN	Auc	0.501	0.7397	0.7434	0.7344	+46.6%
Gemma3-conversation	COMPGCN	Mrr	0.7302	0.9267	0.9301	0.9295	+27.3%
Gemma3-conversation	COMPGCN	HIT@RATE@5	0.8815	1	0.9983	1	+13.4%

Gemma3-conversation	COMPGCN	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-conversation	COMPGCN	Map@5	0.6392	0.9055	0.9062	0.9059	+41.7%
Gemma3-conversation	COMPGCN	Map@10	0.6077	0.8861	0.8865	0.8854	+45.7%
Gemma3-conversation	DEEPPFM	Accuracy	0.5055	0.685	0.6769	0.6817	+34.9%
Gemma3-conversation	DEEPPFM	Precision@5	0.4898	0.8565	0.8542	0.8545	+74.5%
Gemma3-conversation	DEEPPFM	Precision@10	0.4899	0.8365	0.8412	0.8392	+71.3%
Gemma3-conversation	DEEPPFM	Recall@5	0.7158	0.5644	0.564	0.564	-21.20%
Gemma3-conversation	DEEPPFM	Recall@10	0.8815	0.6253	0.6255	0.6253	-29.10%
Gemma3-conversation	DEEPPFM	Ndcg@5	0.7007	0.8596	0.8583	0.8592	+22.6%
Gemma3-conversation	DEEPPFM	Ndcg@10	0.7847	0.8481	0.8513	0.8504	+8.4%
Gemma3-conversation	DEEPPFM	Auc	0.5047	0.733	0.7251	0.726	+43.9%
Gemma3-conversation	DEEPPFM	Mrr	0.7297	0.9311	0.9293	0.9296	+27.4%
Gemma3-conversation	DEEPPFM	HIT@RATE@5	0.8828	1	1	1	+13.3%
Gemma3-conversation	DEEPPFM	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-conversation	DEEPPFM	Map@5	0.643	0.9088	0.9078	0.9107	+41.6%
Gemma3-conversation	DEEPPFM	Map@10	0.6075	0.8875	0.8876	0.8887	+46.3%
Gemma3-conversation	GAT	Accuracy	0.5069	0.6875	0.6828	0.6793	+34.0%
Gemma3-conversation	GAT	Precision@5	0.4881	0.8598	0.8539	0.8539	+74.9%
Gemma3-conversation	GAT	Precision@10	0.4896	0.8398	0.8385	0.8367	+70.9%
Gemma3-conversation	GAT	Recall@5	0.715	0.5643	0.564	0.5641	-21.10%
Gemma3-conversation	GAT	Recall@10	0.8811	0.6255	0.6253	0.6256	-29.00%
Gemma3-conversation	GAT	Ndcg@5	0.698	0.8657	0.858	0.8584	+23.0%
Gemma3-conversation	GAT	Ndcg@10	0.7827	0.853	0.8493	0.8484	+8.4%
Gemma3-conversation	GAT	Auc	0.4989	0.739	0.7319	0.7291	+46.1%
Gemma3-conversation	GAT	Mrr	0.7279	0.9406	0.9285	0.9296	+27.7%
Gemma3-conversation	GAT	HIT@RATE@5	0.884	1	1	0.9983	+12.9%
Gemma3-conversation	GAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	GAT	Map@5	0.643	0.915	0.9081	0.9089	+41.3%
Gemma3-conversation	GAT	Map@10	0.6052	0.8909	0.8875	0.8869	+46.5%
Gemma3-conversation	GCN	Accuracy	0.5069	0.6812	0.6788	0.6739	+32.9%
Gemma3-conversation	GCN	Precision@5	0.4901	0.8582	0.8575	0.8536	+74.2%
Gemma3-conversation	GCN	Precision@10	0.4899	0.8382	0.8408	0.8352	+70.5%
Gemma3-conversation	GCN	Recall@5	0.7181	0.5644	0.5639	0.5638	-21.50%
Gemma3-conversation	GCN	Recall@10	0.8817	0.6253	0.6257	0.6247	-29.10%
Gemma3-conversation	GCN	Ndcg@5	0.7019	0.8627	0.8597	0.8603	+22.6%
Gemma3-conversation	GCN	Ndcg@10	0.7852	0.8504	0.8508	0.8486	+8.1%
Gemma3-conversation	GCN	Auc	0.5027	0.7317	0.722	0.7189	+43.0%
Gemma3-conversation	GCN	Mrr	0.7307	0.9329	0.9254	0.9403	+28.7%
Gemma3-conversation	GCN	HIT@RATE@5	0.8855	1	0.9967	1	+12.9%
Gemma3-conversation	GCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	GCN	Map@5	0.6434	0.9116	0.9074	0.9114	+41.7%
Gemma3-conversation	GCN	Map@10	0.6171	0.8885	0.8889	0.8871	+43.8%
Gemma3-conversation	GRAPHSAGE	Accuracy	0.6908	0.6803	0.6819	0.6718	-2.80%
Gemma3-conversation	GRAPHSAGE	Precision@5	0.7639	0.8572	0.8526	0.8502	+11.3%
Gemma3-conversation	GRAPHSAGE	Precision@10	0.7393	0.8412	0.8339	0.8364	+13.1%
Gemma3-conversation	GRAPHSAGE	Recall@5	0.4333	0.5636	0.5638	0.564	+30.2%
Gemma3-conversation	GRAPHSAGE	Recall@10	0.5326	0.6254	0.6246	0.6254	+17.4%
Gemma3-conversation	GRAPHSAGE	Ndcg@5	0.813	0.8583	0.855	0.8525	+4.9%
Gemma3-conversation	GRAPHSAGE	Ndcg@10	0.8052	0.85	0.8446	0.8455	+5.0%
Gemma3-conversation	GRAPHSAGE	Auc	0.5	0.7311	0.727	0.7194	+43.9%
Gemma3-conversation	GRAPHSAGE	Mrr	0.5	0.9209	0.9241	0.922	+84.4%
Gemma3-conversation	GRAPHSAGE	HIT@RATE@5	0.9931	0.9967	0.9983	0.9983	+0.5%
Gemma3-conversation	GRAPHSAGE	HIT@RATE@10	0.9976	1	1	1	+0.2%
Gemma3-conversation	GRAPHSAGE	Map@5	0.7639	0.9062	0.9036	0.903	+18.2%
Gemma3-conversation	GRAPHSAGE	Map@10	0.7393	0.8865	0.8849	0.8827	+19.4%
Gemma3-conversation	LIGHTGCN	Accuracy	0.5076	0.6806	0.68	0.6768	+33.3%
Gemma3-conversation	LIGHTGCN	Precision@5	0.491	0.8572	0.8512	0.8572	+74.6%
Gemma3-conversation	LIGHTGCN	Precision@10	0.4896	0.8375	0.837	0.8364	+70.8%
Gemma3-conversation	LIGHTGCN	Recall@5	0.7187	0.564	0.5642	0.564	-21.50%

Gemma3-conversation	LIGHTGCN	Recall@10	0.8809	0.6258	0.6258	0.6254	-29.00%
Gemma3-conversation	LIGHTGCN	Ndcg@5	0.7051	0.8598	0.8559	0.8594	+21.9%
Gemma3-conversation	LIGHTGCN	Ndcg@10	0.7871	0.8486	0.8477	0.8474	+7.7%
Gemma3-conversation	LIGHTGCN	Auc	0.5013	0.7302	0.7276	0.7218	+44.0%
Gemma3-conversation	LIGHTGCN	Mrr	0.7348	0.9263	0.9292	0.9307	+26.7%
Gemma3-conversation	LIGHTGCN	HIT@RATE@5	0.8855	1	0.9983	1	+12.9%
Gemma3-conversation	LIGHTGCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	LIGHTGCN	Map@5	0.647	0.9084	0.9069	0.906	+40.0%
Gemma3-conversation	LIGHTGCN	Map@10	0.6108	0.8875	0.8844	0.8851	+44.9%
Gemma3-conversation	KGAT	Accuracy	0.5069	0.6866	0.6802	0.6864	+35.4%
Gemma3-conversation	KGAT	Precision@5	0.4881	0.8575	0.8565	0.8555	+75.3%
Gemma3-conversation	KGAT	Precision@10	0.4896	0.8377	0.8349	0.8397	+71.5%
Gemma3-conversation	KGAT	Recall@5	0.715	0.5641	0.5641	0.5635	-21.20%
Gemma3-conversation	KGAT	Recall@10	0.8811	0.6252	0.6254	0.6258	-29.00%
Gemma3-conversation	KGAT	Ndcg@5	0.698	0.86	0.8597	0.8593	+23.1%
Gemma3-conversation	KGAT	Ndcg@10	0.7827	0.8488	0.8469	0.8504	+8.6%
Gemma3-conversation	KGAT	Auc	0.4989	0.7378	0.7274	0.737	+47.7%
Gemma3-conversation	KGAT	Mrr	0.7279	0.9274	0.9298	0.9316	+28.0%
Gemma3-conversation	KGAT	HIT@RATE@5	0.884	0.9983	0.9983	1	+13.1%
Gemma3-conversation	KGAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	KGAT	Map@5	0.643	0.9064	0.9069	0.9089	+41.4%
Gemma3-conversation	KGAT	Map@10	0.6052	0.8867	0.8856	0.8876	+46.7%
Gemma3-conversation	KGIN	Accuracy	0.5069	0.6885	0.6844	0.6847	+35.1%
Gemma3-conversation	KGIN	Precision@5	0.4906	0.8545	0.8555	0.8569	+74.7%
Gemma3-conversation	KGIN	Precision@10	0.4899	0.8377	0.8408	0.837	+70.9%
Gemma3-conversation	KGIN	Recall@5	0.7181	0.5637	0.5641	0.564	-21.50%
Gemma3-conversation	KGIN	Recall@10	0.8816	0.625	0.6264	0.6256	-29.00%
Gemma3-conversation	KGIN	Ndcg@5	0.7018	0.8581	0.8601	0.861	+22.7%
Gemma3-conversation	KGIN	Ndcg@10	0.7848	0.8487	0.8518	0.849	+8.2%
Gemma3-conversation	KGIN	Auc	0.5054	0.7391	0.7344	0.729	+44.2%
Gemma3-conversation	KGIN	Mrr	0.7305	0.9327	0.9352	0.9337	+27.8%
Gemma3-conversation	KGIN	HIT@RATE@5	0.8847	1	1	0.9983	+12.8%
Gemma3-conversation	KGIN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	KGIN	Map@5	0.6462	0.9056	0.9073	0.9095	+40.7%
Gemma3-conversation	KGIN	Map@10	0.6161	0.8866	0.8872	0.8867	+43.9%
Gemma3-conversation	MULTIGCCF	Accuracy	0.5066	0.6887	0.6876	0.6845	+35.1%
Gemma3-conversation	MULTIGCCF	Precision@5	0.4879	0.8539	0.8502	0.8536	+74.9%
Gemma3-conversation	MULTIGCCF	Precision@10	0.4897	0.8372	0.8388	0.8355	+70.6%
Gemma3-conversation	MULTIGCCF	Recall@5	0.7139	0.5637	0.5635	0.5639	-21.00%
Gemma3-conversation	MULTIGCCF	Recall@10	0.8811	0.6253	0.6256	0.625	-29.10%
Gemma3-conversation	MULTIGCCF	Ndcg@5	0.6979	0.8582	0.856	0.8582	+23.0%
Gemma3-conversation	MULTIGCCF	Ndcg@10	0.7831	0.8485	0.8496	0.8476	+8.2%
Gemma3-conversation	MULTIGCCF	Auc	0.5005	0.7402	0.7376	0.7317	+46.2%
Gemma3-conversation	MULTIGCCF	Mrr	0.7293	0.9296	0.9311	0.9346	+28.1%
Gemma3-conversation	MULTIGCCF	HIT@RATE@5	0.8823	0.9983	1	0.9983	+13.2%
Gemma3-conversation	MULTIGCCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-conversation	MULTIGCCF	Map@5	0.636	0.9089	0.9103	0.9082	+42.8%
Gemma3-conversation	MULTIGCCF	Map@10	0.6064	0.8872	0.8866	0.8867	+46.2%
Gemma3-conversation	NEUMF	Accuracy	0.5069	0.6811	0.678	0.6793	+34.0%
Gemma3-conversation	NEUMF	Precision@5	0.4917	0.8529	0.8516	0.8582	+74.5%
Gemma3-conversation	NEUMF	Precision@10	0.4901	0.8355	0.8344	0.8407	+71.5%
Gemma3-conversation	NEUMF	Recall@5	0.7193	0.5639	0.5638	0.5641	-21.60%
Gemma3-conversation	NEUMF	Recall@10	0.8818	0.6253	0.6252	0.6258	-29.00%
Gemma3-conversation	NEUMF	Ndcg@5	0.7068	0.8547	0.8549	0.8607	+21.8%
Gemma3-conversation	NEUMF	Ndcg@10	0.7892	0.8455	0.8454	0.8512	+7.9%
Gemma3-conversation	NEUMF	Auc	0.5053	0.728	0.7236	0.7233	+43.1%
Gemma3-conversation	NEUMF	Mrr	0.7369	0.9216	0.9276	0.9268	+25.8%
Gemma3-conversation	NEUMF	HIT@RATE@5	0.8826	0.9983	1	0.9983	+13.1%
Gemma3-conversation	NEUMF	HIT@RATE@10	0.8952	1	1	1	+11.7%

Gemma3-conversation	NEUMF	Map@5	0.6509	0.9038	0.906	0.9083	+39.6%
Gemma3-conversation	NEUMF	Map@10	0.6133	0.884	0.8836	0.8887	+44.9%
Gemma3-conversation	NGCF	Accuracy	0.5066	0.689	0.6848	0.6887	+35.9%
Gemma3-conversation	NGCF	Precision@5	0.4879	0.8565	0.8555	0.8555	+75.4%
Gemma3-conversation	NGCF	Precision@10	0.4897	0.8385	0.838	0.8398	+71.5%
Gemma3-conversation	NGCF	Recall@5	0.7139	0.5647	0.5642	0.564	-21.00%
Gemma3-conversation	NGCF	Recall@10	0.8811	0.6261	0.6256	0.6258	-29.00%
Gemma3-conversation	NGCF	Ndcg@5	0.6979	0.8581	0.8586	0.857	+22.8%
Gemma3-conversation	NGCF	Ndcg@10	0.7831	0.8482	0.8486	0.849	+8.4%
Gemma3-conversation	NGCF	Auc	0.5005	0.7411	0.7343	0.7352	+46.9%
Gemma3-conversation	NGCF	Mrr	0.7293	0.9275	0.928	0.925	+26.8%
Gemma3-conversation	NGCF	HIT@RATE@5	0.8823	0.9983	1	1	+13.3%
Gemma3-conversation	NGCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-conversation	NGCF	Map@5	0.636	0.9039	0.9088	0.9042	+42.2%
Gemma3-conversation	NGCF	Map@10	0.6064	0.8846	0.886	0.8853	+46.0%
Gemma3-conversation	PINSAGE	Accuracy	0.5069	0.6888	0.6853	0.6875	+35.6%
Gemma3-conversation	PINSAGE	Precision@5	0.4906	0.8579	0.8559	0.8569	+74.7%
Gemma3-conversation	PINSAGE	Precision@10	0.4899	0.834	0.8317	0.8374	+70.9%
Gemma3-conversation	PINSAGE	Recall@5	0.7181	0.5644	0.5641	0.5641	-21.40%
Gemma3-conversation	PINSAGE	Recall@10	0.8816	0.6247	0.6246	0.6257	-29.00%
Gemma3-conversation	PINSAGE	Ndcg@5	0.7018	0.86	0.8611	0.8612	+22.7%
Gemma3-conversation	PINSAGE	Ndcg@10	0.7848	0.8462	0.8462	0.8497	+8.3%
Gemma3-conversation	PINSAGE	Auc	0.5054	0.7391	0.7337	0.7368	+45.8%
Gemma3-conversation	PINSAGE	Mrr	0.7305	0.9274	0.9342	0.935	+28.0%
Gemma3-conversation	PINSAGE	HIT@RATE@5	0.8847	0.9983	1	0.9983	+12.8%
Gemma3-conversation	PINSAGE	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	PINSAGE	Map@5	0.6462	0.9067	0.9099	0.9091	+40.7%
Gemma3-conversation	PINSAGE	Map@10	0.6161	0.8864	0.8879	0.8882	+44.2%
Gemma3-conversation	SASREC	Accuracy	0.5069	0.6787	0.6802	0.6826	+34.7%
Gemma3-conversation	SASREC	Precision@5	0.4907	0.8479	0.8532	0.8569	+74.6%
Gemma3-conversation	SASREC	Precision@10	0.4895	0.837	0.8375	0.8426	+72.1%
Gemma3-conversation	SASREC	Recall@5	0.7185	0.5637	0.564	0.5644	-21.50%
Gemma3-conversation	SASREC	Recall@10	0.8809	0.6256	0.6254	0.6263	-28.90%
Gemma3-conversation	SASREC	Ndcg@5	0.7048	0.8531	0.8562	0.8593	+21.9%
Gemma3-conversation	SASREC	Ndcg@10	0.7872	0.8471	0.8475	0.852	+8.2%
Gemma3-conversation	SASREC	Auc	0.4981	0.7244	0.7277	0.7307	+46.7%
Gemma3-conversation	SASREC	Mrr	0.7351	0.9299	0.9285	0.9296	+26.5%
Gemma3-conversation	SASREC	HIT@RATE@5	0.887	1	1	1	+12.7%
Gemma3-conversation	SASREC	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	SASREC	Map@5	0.6446	0.9057	0.9049	0.9047	+40.4%
Gemma3-conversation	SASREC	Map@10	0.6096	0.8836	0.8851	0.8873	+45.6%
Gemma3-conversation	SRGNN	Accuracy	0.5081	0.6884	0.6864	0.6804	+33.9%
Gemma3-conversation	SRGNN	Precision@5	0.4896	0.8579	0.8602	0.8549	+74.6%
Gemma3-conversation	SRGNN	Precision@10	0.4898	0.836	0.8332	0.8316	+69.8%
Gemma3-conversation	SRGNN	Recall@5	0.716	0.5641	0.5645	0.5644	-21.20%
Gemma3-conversation	SRGNN	Recall@10	0.8815	0.6251	0.6253	0.6248	-29.10%
Gemma3-conversation	SRGNN	Ndcg@5	0.7006	0.8592	0.8632	0.8578	+22.4%
Gemma3-conversation	SRGNN	Ndcg@10	0.7845	0.847	0.8468	0.844	+7.6%
Gemma3-conversation	SRGNN	Auc	0.5021	0.7391	0.733	0.7269	+44.8%
Gemma3-conversation	SRGNN	Mrr	0.7334	0.9274	0.9322	0.9252	+26.2%
Gemma3-conversation	SRGNN	HIT@RATE@5	0.8838	0.9983	0.9983	0.9983	+13.0%
Gemma3-conversation	SRGNN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-conversation	SRGNN	Map@5	0.6472	0.9049	0.9096	0.9056	+39.9%
Gemma3-conversation	SRGNN	Map@10	0.6192	0.887	0.8878	0.8853	+43.0%
Gemma3-structured	BPR	Accuracy	0.7098	0.6741	0.6545	0.6587	-7.20%
Gemma3-structured	BPR	Precision@5	0.7826	0.8516	0.8274	0.8407	+7.4%
Gemma3-structured	BPR	Precision@10	0.7591	0.8302	0.8104	0.8152	+7.4%
Gemma3-structured	BPR	Recall@5	0.4386	0.5633	0.5623	0.564	+28.6%
Gemma3-structured	BPR	Recall@10	0.544	0.6237	0.6229	0.6233	+14.6%

Gemma3-structured	BPR	Ndcg@5	0.8322	0.8555	0.8325	0.8456	+1.6%
Gemma3-structured	BPR	Ndcg@10	0.8284	0.8425	0.8224	0.8294	+0.1%
Gemma3-structured	BPR	Auc	0.5	0.7166	0.6851	0.6898	+38.0%
Gemma3-structured	BPR	Mrr	0.5	0.9295	0.9115	0.9252	+85.0%
Gemma3-structured	BPR	HIT@RATE@5	0.9938	0.9983	0.9967	1	+0.6%
Gemma3-structured	BPR	HIT@RATE@10	0.9987	1	1	1	+0.1%
Gemma3-structured	BPR	Map@5	0.7826	0.9064	0.8917	0.8998	+15.0%
Gemma3-structured	BPR	Map@10	0.7591	0.8839	0.8662	0.8745	+15.2%
Gemma3-structured	COMPGCN	Accuracy	0.5069	0.6856	0.6868	0.6857	+35.3%
Gemma3-structured	COMPGCN	Precision@5	0.4887	0.8476	0.8569	0.8526	+74.5%
Gemma3-structured	COMPGCN	Precision@10	0.4899	0.8364	0.8354	0.834	+70.2%
Gemma3-structured	COMPGCN	Recall@5	0.7148	0.5631	0.564	0.564	-21.10%
Gemma3-structured	COMPGCN	Recall@10	0.8816	0.6253	0.6249	0.6251	-29.10%
Gemma3-structured	COMPGCN	Ndcg@5	0.6996	0.8521	0.8594	0.8549	+22.2%
Gemma3-structured	COMPGCN	Ndcg@10	0.7845	0.8462	0.8472	0.8444	+7.6%
Gemma3-structured	COMPGCN	Auc	0.501	0.7401	0.7385	0.7318	+46.1%
Gemma3-structured	COMPGCN	Mrr	0.7302	0.9272	0.9293	0.9278	+27.1%
Gemma3-structured	COMPGCN	HIT@RATE@5	0.8815	1	0.9983	0.9983	+13.3%
Gemma3-structured	COMPGCN	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-structured	COMPGCN	Map@5	0.6392	0.9033	0.9055	0.9027	+41.2%
Gemma3-structured	COMPGCN	Map@10	0.6077	0.8827	0.886	0.8825	+45.2%
Gemma3-structured	DEEPPFM	Accuracy	0.5055	0.6879	0.6779	0.6845	+35.4%
Gemma3-structured	DEEPPFM	Precision@5	0.4898	0.8549	0.8522	0.8598	+75.5%
Gemma3-structured	DEEPPFM	Precision@10	0.4899	0.8385	0.8377	0.8355	+70.6%
Gemma3-structured	DEEPPFM	Recall@5	0.7158	0.564	0.5634	0.5643	-21.20%
Gemma3-structured	DEEPPFM	Recall@10	0.8815	0.626	0.6252	0.6253	-29.10%
Gemma3-structured	DEEPPFM	Ndcg@5	0.7007	0.8567	0.8553	0.8616	+23.0%
Gemma3-structured	DEEPPFM	Ndcg@10	0.7847	0.848	0.8477	0.8472	+8.0%
Gemma3-structured	DEEPPFM	Auc	0.5047	0.7351	0.7251	0.7313	+44.9%
Gemma3-structured	DEEPPFM	Mrr	0.7297	0.9227	0.925	0.9291	+27.3%
Gemma3-structured	DEEPPFM	HIT@RATE@5	0.8828	0.9983	1	1	+13.3%
Gemma3-structured	DEEPPFM	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-structured	DEEPPFM	Map@5	0.643	0.9059	0.9046	0.9074	+41.1%
Gemma3-structured	DEEPPFM	Map@10	0.6075	0.8852	0.8858	0.8862	+45.9%
Gemma3-structured	GAT	Accuracy	0.5069	0.688	0.6782	0.6784	+33.8%
Gemma3-structured	GAT	Precision@5	0.4881	0.8575	0.8605	0.8598	+76.2%
Gemma3-structured	GAT	Precision@10	0.4896	0.8357	0.8403	0.838	+71.2%
Gemma3-structured	GAT	Recall@5	0.715	0.5641	0.5641	0.5642	-21.10%
Gemma3-structured	GAT	Recall@10	0.8811	0.6246	0.6257	0.6255	-29.00%
Gemma3-structured	GAT	Ndcg@5	0.698	0.8591	0.8639	0.8616	+23.4%
Gemma3-structured	GAT	Ndcg@10	0.7827	0.8468	0.8523	0.8493	+8.5%
Gemma3-structured	GAT	Auc	0.4989	0.7388	0.7273	0.7264	+45.6%
Gemma3-structured	GAT	Mrr	0.7279	0.926	0.9315	0.931	+27.9%
Gemma3-structured	GAT	HIT@RATE@5	0.884	0.9983	1	1	+13.1%
Gemma3-structured	GAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-structured	GAT	Map@5	0.643	0.9059	0.912	0.9075	+41.1%
Gemma3-structured	GAT	Map@10	0.6052	0.8871	0.891	0.8883	+46.8%
Gemma3-structured	GCN	Accuracy	0.5069	0.6814	0.6818	0.6756	+33.3%
Gemma3-structured	GCN	Precision@5	0.4901	0.8579	0.8522	0.8555	+74.6%
Gemma3-structured	GCN	Precision@10	0.4899	0.8372	0.8393	0.8347	+70.4%
Gemma3-structured	GCN	Recall@5	0.7181	0.5638	0.5639	0.5642	-21.40%
Gemma3-structured	GCN	Recall@10	0.8817	0.6253	0.6254	0.6244	-29.20%
Gemma3-structured	GCN	Ndcg@5	0.7019	0.862	0.8549	0.8611	+22.7%
Gemma3-structured	GCN	Ndcg@10	0.7852	0.8495	0.8485	0.8481	+8.0%
Gemma3-structured	GCN	Auc	0.5027	0.7275	0.7234	0.7194	+43.1%
Gemma3-structured	GCN	Mrr	0.7307	0.9341	0.9242	0.9334	+27.7%
Gemma3-structured	GCN	HIT@RATE@5	0.8855	0.9983	0.9983	0.9983	+12.7%
Gemma3-structured	GCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-structured	GCN	Map@5	0.6434	0.9091	0.9036	0.912	+41.7%

Gemma3-structured	GCN	Map@10	0.6171	0.8887	0.8847	0.8885	+44.0%
Gemma3-structured	GRAPHSAGE	Accuracy	0.6908	0.6819	0.6829	0.6783	-1.80%
Gemma3-structured	GRAPHSAGE	Precision@5	0.7639	0.8605	0.8526	0.8542	+11.8%
Gemma3-structured	GRAPHSAGE	Precision@10	0.7393	0.8377	0.837	0.8352	+13.0%
Gemma3-structured	GRAPHSAGE	Recall@5	0.4333	0.5643	0.5641	0.5639	+30.1%
Gemma3-structured	GRAPHSAGE	Recall@10	0.5326	0.625	0.6257	0.6245	+17.2%
Gemma3-structured	GRAPHSAGE	Ndcg@5	0.813	0.8617	0.8537	0.8566	+5.4%
Gemma3-structured	GRAPHSAGE	Ndcg@10	0.8052	0.8486	0.8457	0.8462	+5.1%
Gemma3-structured	GRAPHSAGE	Auc	0.5	0.7296	0.727	0.7271	+45.4%
Gemma3-structured	GRAPHSAGE	Mrr	0.5	0.9286	0.918	0.9259	+85.2%
Gemma3-structured	GRAPHSAGE	HIT@RATE@5	0.9931	1	1	1	+0.7%
Gemma3-structured	GRAPHSAGE	HIT@RATE@10	0.9976	1	1	1	+0.2%
Gemma3-structured	GRAPHSAGE	Map@5	0.7639	0.9055	0.9016	0.9047	+18.4%
Gemma3-structured	GRAPHSAGE	Map@10	0.7393	0.8866	0.8821	0.8855	+19.8%
Gemma3-structured	LIGHTGCN	Accuracy	0.5076	0.6811	0.6808	0.6771	+33.4%
Gemma3-structured	LIGHTGCN	Precision@5	0.491	0.8579	0.8539	0.8516	+73.4%
Gemma3-structured	LIGHTGCN	Precision@10	0.4896	0.8374	0.8349	0.8309	+69.7%
Gemma3-structured	LIGHTGCN	Recall@5	0.7187	0.564	0.5637	0.5638	-21.60%
Gemma3-structured	LIGHTGCN	Recall@10	0.8809	0.6249	0.6247	0.6249	-29.10%
Gemma3-structured	LIGHTGCN	Ndcg@5	0.7051	0.8619	0.8594	0.8561	+21.4%
Gemma3-structured	LIGHTGCN	Ndcg@10	0.7871	0.8495	0.8475	0.8439	+7.2%
Gemma3-structured	LIGHTGCN	Auc	0.5013	0.7302	0.7233	0.7264	+44.9%
Gemma3-structured	LIGHTGCN	Mrr	0.7348	0.9362	0.9335	0.9272	+26.2%
Gemma3-structured	LIGHTGCN	HIT@RATE@5	0.8855	1	0.9983	1	+12.9%
Gemma3-structured	LIGHTGCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-structured	LIGHTGCN	Map@5	0.647	0.9099	0.9106	0.9073	+40.2%
Gemma3-structured	LIGHTGCN	Map@10	0.6108	0.8875	0.8869	0.8854	+45.0%
Gemma3-structured	KGAT	Accuracy	0.5069	0.6862	0.6851	0.6817	+34.5%
Gemma3-structured	KGAT	Precision@5	0.4881	0.8555	0.8545	0.8522	+74.6%
Gemma3-structured	KGAT	Precision@10	0.4896	0.8426	0.8369	0.834	+70.4%
Gemma3-structured	KGAT	Recall@5	0.715	0.5641	0.5636	0.5643	-21.10%
Gemma3-structured	KGAT	Recall@10	0.8811	0.626	0.625	0.6253	-29.00%
Gemma3-structured	KGAT	Ndcg@5	0.698	0.8581	0.8555	0.8577	+22.9%
Gemma3-structured	KGAT	Ndcg@10	0.7827	0.8516	0.8462	0.8463	+8.1%
Gemma3-structured	KGAT	Auc	0.4989	0.7386	0.7287	0.7313	+46.6%
Gemma3-structured	KGAT	Mrr	0.7279	0.9271	0.9238	0.9341	+28.3%
Gemma3-structured	KGAT	HIT@RATE@5	0.884	0.9983	1	0.9967	+12.7%
Gemma3-structured	KGAT	HIT@RATE@10	0.8954	1	1	0.9983	+11.5%
Gemma3-structured	KGAT	Map@5	0.643	0.9055	0.9026	0.9066	+41.0%
Gemma3-structured	KGAT	Map@10	0.6052	0.8871	0.8837	0.8835	+46.0%
Gemma3-structured	KGIN	Accuracy	0.5069	0.6886	0.6876	0.6873	+35.6%
Gemma3-structured	KGIN	Precision@5	0.4906	0.8595	0.8549	0.8582	+74.9%
Gemma3-structured	KGIN	Precision@10	0.4899	0.8383	0.8369	0.8382	+71.1%
Gemma3-structured	KGIN	Recall@5	0.7181	0.5646	0.5642	0.5641	-21.40%
Gemma3-structured	KGIN	Recall@10	0.8816	0.6248	0.6258	0.6254	-29.10%
Gemma3-structured	KGIN	Ndcg@5	0.7018	0.8637	0.8598	0.8612	+22.7%
Gemma3-structured	KGIN	Ndcg@10	0.7848	0.8512	0.8491	0.8496	+8.3%
Gemma3-structured	KGIN	Auc	0.5054	0.7373	0.7361	0.7342	+45.3%
Gemma3-structured	KGIN	Mrr	0.7305	0.9358	0.9333	0.9277	+27.0%
Gemma3-structured	KGIN	HIT@RATE@5	0.8847	1	1	1	+13.0%
Gemma3-structured	KGIN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-structured	KGIN	Map@5	0.6462	0.9101	0.9101	0.9095	+40.7%
Gemma3-structured	KGIN	Map@10	0.6161	0.8905	0.8875	0.8887	+44.2%
Gemma3-structured	MULTIGCCF	Accuracy	0.5066	0.6883	0.6846	0.6888	+36.0%
Gemma3-structured	MULTIGCCF	Precision@5	0.4879	0.8562	0.8598	0.8542	+75.1%
Gemma3-structured	MULTIGCCF	Precision@10	0.4897	0.8354	0.8362	0.8377	+71.1%
Gemma3-structured	MULTIGCCF	Recall@5	0.7139	0.5638	0.5642	0.5638	-21.00%
Gemma3-structured	MULTIGCCF	Recall@10	0.8811	0.6248	0.6254	0.6254	-29.00%
Gemma3-structured	MULTIGCCF	Ndcg@5	0.6979	0.8598	0.864	0.8579	+22.9%

Gemma3-structured	MULTIGCCF	Ndcg@10	0.7831	0.8476	0.8495	0.8485	+8.4%
Gemma3-structured	MULTIGCCF	Auc	0.5005	0.7399	0.7331	0.7386	+47.6%
Gemma3-structured	MULTIGCCF	Mrr	0.7293	0.9314	0.939	0.9288	+27.4%
Gemma3-structured	MULTIGCCF	HIT@RATE@5	0.8823	0.9983	1	0.9983	+13.2%
Gemma3-structured	MULTIGCCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-structured	MULTIGCCF	Map@5	0.636	0.9077	0.9102	0.9075	+42.7%
Gemma3-structured	MULTIGCCF	Map@10	0.6064	0.8876	0.8886	0.8865	+46.2%
Gemma3-structured	NEUMF	Accuracy	0.5069	0.6818	0.6831	0.6772	+33.6%
Gemma3-structured	NEUMF	Precision@5	0.4917	0.8549	0.8575	0.8569	+74.3%
Gemma3-structured	NEUMF	Precision@10	0.4901	0.8349	0.8379	0.8352	+70.4%
Gemma3-structured	NEUMF	Recall@5	0.7193	0.564	0.5641	0.5642	-21.60%
Gemma3-structured	NEUMF	Recall@10	0.8818	0.6249	0.6251	0.6246	-29.20%
Gemma3-structured	NEUMF	Ndcg@5	0.7068	0.8579	0.8643	0.8585	+21.5%
Gemma3-structured	NEUMF	Ndcg@10	0.7892	0.8464	0.8515	0.8466	+7.3%
Gemma3-structured	NEUMF	Auc	0.5053	0.7311	0.7312	0.7222	+42.9%
Gemma3-structured	NEUMF	Mrr	0.7369	0.9275	0.9404	0.926	+25.7%
Gemma3-structured	NEUMF	HIT@RATE@5	0.8826	0.9967	0.9983	1	+13.3%
Gemma3-structured	NEUMF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-structured	NEUMF	Map@5	0.6509	0.9052	0.9151	0.9067	+39.3%
Gemma3-structured	NEUMF	Map@10	0.6133	0.8858	0.891	0.887	+44.6%
Gemma3-structured	NGCF	Accuracy	0.5066	0.6883	0.6855	0.6859	+35.4%
Gemma3-structured	NGCF	Precision@5	0.4879	0.8536	0.8529	0.8575	+75.8%
Gemma3-structured	NGCF	Precision@10	0.4897	0.8385	0.8369	0.8375	+71.0%
Gemma3-structured	NGCF	Recall@5	0.7139	0.5635	0.5639	0.564	-21.00%
Gemma3-structured	NGCF	Recall@10	0.8811	0.6248	0.6248	0.6256	-29.00%
Gemma3-structured	NGCF	Ndcg@5	0.6979	0.8606	0.8564	0.8581	+23.0%
Gemma3-structured	NGCF	Ndcg@10	0.7831	0.8511	0.8474	0.8476	+8.2%
Gemma3-structured	NGCF	Auc	0.5005	0.7392	0.7332	0.7348	+46.8%
Gemma3-structured	NGCF	Mrr	0.7293	0.9419	0.9276	0.9253	+26.9%
Gemma3-structured	NGCF	HIT@RATE@5	0.8823	1	1	1	+13.3%
Gemma3-structured	NGCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Gemma3-structured	NGCF	Map@5	0.636	0.9117	0.9072	0.9036	+42.1%
Gemma3-structured	NGCF	Map@10	0.6064	0.8879	0.8857	0.8855	+46.0%
Gemma3-structured	PINSAGE	Accuracy	0.5069	0.6889	0.6836	0.6866	+35.5%
Gemma3-structured	PINSAGE	Precision@5	0.4906	0.8615	0.8592	0.8539	+74.0%
Gemma3-structured	PINSAGE	Precision@10	0.4899	0.842	0.8369	0.8347	+70.4%
Gemma3-structured	PINSAGE	Recall@5	0.7181	0.5644	0.5648	0.5648	-21.40%
Gemma3-structured	PINSAGE	Recall@10	0.8816	0.6264	0.6255	0.6252	-29.10%
Gemma3-structured	PINSAGE	Ndcg@5	0.7018	0.8634	0.8614	0.8597	+22.5%
Gemma3-structured	PINSAGE	Ndcg@10	0.7848	0.8524	0.8487	0.8478	+8.0%
Gemma3-structured	PINSAGE	Auc	0.5054	0.7399	0.7301	0.7274	+43.9%
Gemma3-structured	PINSAGE	Mrr	0.7305	0.9328	0.9286	0.9358	+28.1%
Gemma3-structured	PINSAGE	HIT@RATE@5	0.8847	1	1	0.9983	+12.8%
Gemma3-structured	PINSAGE	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-structured	PINSAGE	Map@5	0.6462	0.9086	0.9082	0.91	+40.8%
Gemma3-structured	PINSAGE	Map@10	0.6161	0.8878	0.8887	0.8874	+44.0%
Gemma3-structured	SASREC	Accuracy	0.5069	0.6808	0.6764	0.6746	+33.1%
Gemma3-structured	SASREC	Precision@5	0.4907	0.8545	0.8545	0.8598	+75.2%
Gemma3-structured	SASREC	Precision@10	0.4895	0.8425	0.8392	0.8423	+72.1%
Gemma3-structured	SASREC	Recall@5	0.7185	0.5634	0.5642	0.565	-21.40%
Gemma3-structured	SASREC	Recall@10	0.8809	0.6257	0.6252	0.6261	-28.90%
Gemma3-structured	SASREC	Ndcg@5	0.7048	0.8588	0.8565	0.8599	+22.0%
Gemma3-structured	SASREC	Ndcg@10	0.7872	0.8524	0.8488	0.851	+8.1%
Gemma3-structured	SASREC	Auc	0.4981	0.7335	0.722	0.7164	+43.8%
Gemma3-structured	SASREC	Mrr	0.7351	0.9278	0.9229	0.9239	+25.7%
Gemma3-structured	SASREC	HIT@RATE@5	0.887	0.9983	1	0.9983	+12.6%
Gemma3-structured	SASREC	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-structured	SASREC	Map@5	0.6446	0.91	0.9041	0.9035	+40.2%
Gemma3-structured	SASREC	Map@10	0.6096	0.8884	0.8868	0.8866	+45.4%

Gemma3-structured	SRGNN	Accuracy	0.5081	0.6869	0.6812	0.6731	+32.5%
Gemma3-structured	SRGNN	Precision@5	0.4896	0.8552	0.8545	0.8512	+73.9%
Gemma3-structured	SRGNN	Precision@10	0.4898	0.8392	0.8324	0.8383	+71.2%
Gemma3-structured	SRGNN	Recall@5	0.716	0.5641	0.564	0.5635	-21.30%
Gemma3-structured	SRGNN	Recall@10	0.8815	0.6256	0.6247	0.6247	-29.10%
Gemma3-structured	SRGNN	Ndcg@5	0.7006	0.8609	0.8572	0.855	+22.0%
Gemma3-structured	SRGNN	Ndcg@10	0.7845	0.8511	0.8446	0.8482	+8.1%
Gemma3-structured	SRGNN	Auc	0.5021	0.7353	0.7315	0.7208	+43.6%
Gemma3-structured	SRGNN	Mrr	0.7334	0.9348	0.9275	0.9258	+26.2%
Gemma3-structured	SRGNN	HIT@RATE@5	0.8838	0.9983	0.9967	0.9983	+13.0%
Gemma3-structured	SRGNN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Gemma3-structured	SRGNN	Map@5	0.6472	0.9112	0.9047	0.9051	+39.9%
Gemma3-structured	SRGNN	Map@10	0.6192	0.8877	0.8862	0.8854	+43.0%
Qwen3-complex	BPR	Accuracy	0.7098	0.6824	0.6602	0.6534	-8.00%
Qwen3-complex	BPR	Precision@5	0.7826	0.8493	0.8344	0.8205	+4.8%
Qwen3-complex	BPR	Precision@10	0.7591	0.8302	0.8231	0.8005	+5.5%
Qwen3-complex	BPR	Recall@5	0.4386	0.5635	0.5631	0.5629	+28.3%
Qwen3-complex	BPR	Recall@10	0.544	0.6242	0.6245	0.622	+14.3%
Qwen3-complex	BPR	Ndcg@5	0.8322	0.8544	0.8362	0.8231	-1.10%
Qwen3-complex	BPR	Ndcg@10	0.8284	0.8429	0.8312	0.8116	-2.00%
Qwen3-complex	BPR	Auc	0.5	0.726	0.7	0.6802	+36.0%
Qwen3-complex	BPR	Mrr	0.5	0.933	0.9076	0.9095	+81.9%
Qwen3-complex	BPR	HIT@RATE@5	0.9938	1	1	0.9967	+0.3%
Qwen3-complex	BPR	HIT@RATE@10	0.9987	1	1	0.9983	0.00%
Qwen3-complex	BPR	Map@5	0.7826	0.9051	0.892	0.881	+12.6%
Qwen3-complex	BPR	Map@10	0.7591	0.882	0.8707	0.857	+12.9%
Qwen3-complex	COMPGCN	Accuracy	0.5069	0.6743	0.6873	0.6874	+35.6%
Qwen3-complex	COMPGCN	Precision@5	0.4887	0.8552	0.8519	0.8598	+75.9%
Qwen3-complex	COMPGCN	Precision@10	0.4899	0.8339	0.8388	0.8372	+70.9%
Qwen3-complex	COMPGCN	Recall@5	0.7148	0.5636	0.5638	0.564	-21.10%
Qwen3-complex	COMPGCN	Recall@10	0.8816	0.6245	0.6255	0.6253	-29.10%
Qwen3-complex	COMPGCN	Ndcg@5	0.6996	0.8588	0.8562	0.8612	+23.1%
Qwen3-complex	COMPGCN	Ndcg@10	0.7845	0.8461	0.849	0.8483	+8.1%
Qwen3-complex	COMPGCN	Auc	0.501	0.74	0.7366	0.7385	+47.4%
Qwen3-complex	COMPGCN	Mrr	0.7302	0.9338	0.931	0.9284	+27.1%
Qwen3-complex	COMPGCN	HIT@RATE@5	0.8815	1	0.9967	1	+13.4%
Qwen3-complex	COMPGCN	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-complex	COMPGCN	Map@5	0.6392	0.9058	0.9055	0.9064	+41.8%
Qwen3-complex	COMPGCN	Map@10	0.6077	0.886	0.8854	0.887	+46.0%
Qwen3-complex	DEEPPFM	Accuracy	0.5055	0.683	0.6812	0.6768	+33.9%
Qwen3-complex	DEEPPFM	Precision@5	0.4898	0.8572	0.8552	0.8542	+74.4%
Qwen3-complex	DEEPPFM	Precision@10	0.4899	0.8382	0.8352	0.8362	+70.7%
Qwen3-complex	DEEPPFM	Recall@5	0.7158	0.5646	0.5637	0.5638	-21.20%
Qwen3-complex	DEEPPFM	Recall@10	0.8815	0.6252	0.6254	0.625	-29.10%
Qwen3-complex	DEEPPFM	Ndcg@5	0.7007	0.8599	0.8612	0.8573	+22.3%
Qwen3-complex	DEEPPFM	Ndcg@10	0.7847	0.8488	0.8485	0.8473	+8.0%
Qwen3-complex	DEEPPFM	Auc	0.5047	0.7311	0.7256	0.7192	+42.5%
Qwen3-complex	DEEPPFM	Mrr	0.7297	0.9255	0.9356	0.9283	+27.2%
Qwen3-complex	DEEPPFM	HIT@RATE@5	0.8828	0.9983	1	1	+13.3%
Qwen3-complex	DEEPPFM	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-complex	DEEPPFM	Map@5	0.643	0.9093	0.9119	0.9057	+40.9%
Qwen3-complex	DEEPPFM	Map@10	0.6075	0.8863	0.8874	0.8861	+45.9%
Qwen3-complex	GAT	Accuracy	0.5069	0.6871	0.6759	0.6857	+35.3%
Qwen3-complex	GAT	Precision@5	0.4881	0.8612	0.8549	0.8555	+75.3%
Qwen3-complex	GAT	Precision@10	0.4896	0.8425	0.8367	0.8375	+71.1%
Qwen3-complex	GAT	Recall@5	0.715	0.5643	0.5642	0.5641	-21.10%
Qwen3-complex	GAT	Recall@10	0.8811	0.6264	0.6261	0.6256	-29.00%
Qwen3-complex	GAT	Ndcg@5	0.698	0.8657	0.8595	0.8572	+22.8%
Qwen3-complex	GAT	Ndcg@10	0.7827	0.8545	0.8487	0.8475	+8.3%

Qwen3-complex	GAT	Auc	0.4989	0.7378	0.719	0.7362	+47.6%
Qwen3-complex	GAT	Mrr	0.7279	0.938	0.9344	0.9251	+27.1%
Qwen3-complex	GAT	HIT@RATE@5	0.884	1	0.9983	0.9983	+12.9%
Qwen3-complex	GAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-complex	GAT	Map@5	0.643	0.9115	0.9083	0.9035	+40.5%
Qwen3-complex	GAT	Map@10	0.6052	0.8914	0.8871	0.8849	+46.2%
Qwen3-complex	GCN	Accuracy	0.5069	0.68	0.6807	0.6723	+32.6%
Qwen3-complex	GCN	Precision@5	0.4901	0.8565	0.8565	0.8598	+75.4%
Qwen3-complex	GCN	Precision@10	0.4899	0.8398	0.8379	0.8415	+71.8%
Qwen3-complex	GCN	Recall@5	0.7181	0.5639	0.5641	0.5643	-21.40%
Qwen3-complex	GCN	Recall@10	0.8817	0.6256	0.6254	0.626	-29.00%
Qwen3-complex	GCN	Ndcg@5	0.7019	0.8587	0.8573	0.8643	+23.1%
Qwen3-complex	GCN	Ndcg@10	0.7852	0.8496	0.8475	0.8532	+8.7%
Qwen3-complex	GCN	Auc	0.5027	0.7256	0.727	0.7176	+42.7%
Qwen3-complex	GCN	Mrr	0.7307	0.9277	0.9221	0.9321	+27.6%
Qwen3-complex	GCN	HIT@RATE@5	0.8855	0.9983	0.9983	1	+12.9%
Qwen3-complex	GCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-complex	GCN	Map@5	0.6434	0.9049	0.9026	0.913	+41.9%
Qwen3-complex	GCN	Map@10	0.6171	0.8856	0.8856	0.8901	+44.2%
Qwen3-complex	GRAPHSAGE	Accuracy	0.6908	0.677	0.6775	0.6862	-0.70%
Qwen3-complex	GRAPHSAGE	Precision@5	0.7639	0.8562	0.8565	0.8536	+11.7%
Qwen3-complex	GRAPHSAGE	Precision@10	0.7393	0.8379	0.8355	0.836	+13.1%
Qwen3-complex	GRAPHSAGE	Recall@5	0.4333	0.5639	0.5643	0.5638	+30.1%
Qwen3-complex	GRAPHSAGE	Recall@10	0.5326	0.6252	0.6255	0.6251	+17.4%
Qwen3-complex	GRAPHSAGE	Ndcg@5	0.813	0.8607	0.8594	0.8581	+5.6%
Qwen3-complex	GRAPHSAGE	Ndcg@10	0.8052	0.8496	0.847	0.8477	+5.3%
Qwen3-complex	GRAPHSAGE	Auc	0.5	0.7218	0.7239	0.73	+46.0%
Qwen3-complex	GRAPHSAGE	Mrr	0.5	0.9285	0.9286	0.9321	+86.4%
Qwen3-complex	GRAPHSAGE	HIT@RATE@5	0.9931	0.9983	1	1	+0.7%
Qwen3-complex	GRAPHSAGE	HIT@RATE@10	0.9976	1	1	1	+0.2%
Qwen3-complex	GRAPHSAGE	Map@5	0.7639	0.911	0.9084	0.9076	+18.8%
Qwen3-complex	GRAPHSAGE	Map@10	0.7393	0.8879	0.887	0.8858	+19.8%
Qwen3-complex	LIGHTGCN	Accuracy	0.5076	0.6817	0.6832	0.6813	+34.2%
Qwen3-complex	LIGHTGCN	Precision@5	0.491	0.8509	0.8519	0.8536	+73.8%
Qwen3-complex	LIGHTGCN	Precision@10	0.4896	0.8317	0.8398	0.8372	+71.0%
Qwen3-complex	LIGHTGCN	Recall@5	0.7187	0.5637	0.5631	0.564	-21.50%
Qwen3-complex	LIGHTGCN	Recall@10	0.8809	0.6248	0.6255	0.6255	-29.00%
Qwen3-complex	LIGHTGCN	Ndcg@5	0.7051	0.8556	0.8565	0.8563	+21.5%
Qwen3-complex	LIGHTGCN	Ndcg@10	0.7871	0.8442	0.85	0.8474	+7.7%
Qwen3-complex	LIGHTGCN	Auc	0.5013	0.7218	0.7313	0.7298	+45.6%
Qwen3-complex	LIGHTGCN	Mrr	0.7348	0.9311	0.9293	0.9267	+26.1%
Qwen3-complex	LIGHTGCN	HIT@RATE@5	0.8855	1	1	1	+12.9%
Qwen3-complex	LIGHTGCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-complex	LIGHTGCN	Map@5	0.647	0.9067	0.9075	0.9046	+39.8%
Qwen3-complex	LIGHTGCN	Map@10	0.6108	0.8842	0.8865	0.8839	+44.7%
Qwen3-complex	KGAT	Accuracy	0.5069	0.6874	0.6708	0.6788	+33.9%
Qwen3-complex	KGAT	Precision@5	0.4881	0.8582	0.8496	0.8579	+75.8%
Qwen3-complex	KGAT	Precision@10	0.4896	0.8405	0.8319	0.8364	+70.8%
Qwen3-complex	KGAT	Recall@5	0.715	0.5643	0.5643	0.5644	-21.10%
Qwen3-complex	KGAT	Recall@10	0.8811	0.6258	0.6247	0.6254	-29.00%
Qwen3-complex	KGAT	Ndcg@5	0.698	0.8629	0.8521	0.8625	+23.6%
Qwen3-complex	KGAT	Ndcg@10	0.7827	0.8523	0.8423	0.8491	+8.5%
Qwen3-complex	KGAT	Auc	0.4989	0.7383	0.709	0.7305	+46.4%
Qwen3-complex	KGAT	Mrr	0.7279	0.9354	0.9224	0.9344	+28.4%
Qwen3-complex	KGAT	HIT@RATE@5	0.884	1	0.9934	1	+13.1%
Qwen3-complex	KGAT	HIT@RATE@10	0.8954	1	0.9967	1	+11.7%
Qwen3-complex	KGAT	Map@5	0.643	0.9131	0.9002	0.9117	+41.8%
Qwen3-complex	KGAT	Map@10	0.6052	0.8895	0.8814	0.8883	+46.8%
Qwen3-complex	KGIN	Accuracy	0.5069	0.6896	0.6858	0.6867	+35.5%

Qwen3-complex	KGIN	Precision@5	0.4906	0.8526	0.8559	0.8612	+75.5%
Qwen3-complex	KGIN	Precision@10	0.4899	0.8369	0.8395	0.8387	+71.2%
Qwen3-complex	KGIN	Recall@5	0.7181	0.5638	0.5642	0.5646	-21.40%
Qwen3-complex	KGIN	Recall@10	0.8816	0.6254	0.626	0.6258	-29.00%
Qwen3-complex	KGIN	Ndcg@5	0.7018	0.8576	0.8585	0.8632	+23.0%
Qwen3-complex	KGIN	Ndcg@10	0.7848	0.8484	0.8498	0.8502	+8.3%
Qwen3-complex	KGIN	Auc	0.5054	0.7396	0.734	0.7344	+45.3%
Qwen3-complex	KGIN	Mrr	0.7305	0.9302	0.9295	0.9317	+27.5%
Qwen3-complex	KGIN	HIT@RATE@5	0.8847	0.9967	1	1	+13.0%
Qwen3-complex	KGIN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-complex	KGIN	Map@5	0.6462	0.9081	0.9054	0.9088	+40.6%
Qwen3-complex	KGIN	Map@10	0.6161	0.8875	0.8868	0.8882	+44.2%
Qwen3-complex	MULTIGCCF	Accuracy	0.5066	0.6899	0.6884	0.6842	+35.1%
Qwen3-complex	MULTIGCCF	Precision@5	0.4879	0.8575	0.8592	0.8562	+75.5%
Qwen3-complex	MULTIGCCF	Precision@10	0.4897	0.8359	0.8362	0.8387	+71.3%
Qwen3-complex	MULTIGCCF	Recall@5	0.7139	0.5643	0.5643	0.5641	-21.00%
Qwen3-complex	MULTIGCCF	Recall@10	0.8811	0.6253	0.6253	0.6258	-29.00%
Qwen3-complex	MULTIGCCF	Ndcg@5	0.6979	0.8583	0.8629	0.8586	+23.0%
Qwen3-complex	MULTIGCCF	Ndcg@10	0.7831	0.8466	0.849	0.849	+8.4%
Qwen3-complex	MULTIGCCF	Auc	0.5005	0.7407	0.7386	0.7309	+46.0%
Qwen3-complex	MULTIGCCF	Mrr	0.7293	0.9257	0.9327	0.9284	+27.3%
Qwen3-complex	MULTIGCCF	HIT@RATE@5	0.8823	0.9983	0.9983	1	+13.3%
Qwen3-complex	MULTIGCCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-complex	MULTIGCCF	Map@5	0.636	0.9047	0.9107	0.906	+42.5%
Qwen3-complex	MULTIGCCF	Map@10	0.6064	0.8866	0.888	0.886	+46.1%
Qwen3-complex	NEUMF	Accuracy	0.5069	0.679	0.6707	0.6741	+33.0%
Qwen3-complex	NEUMF	Precision@5	0.4917	0.8529	0.8516	0.8552	+73.9%
Qwen3-complex	NEUMF	Precision@10	0.4901	0.8342	0.8383	0.8322	+69.8%
Qwen3-complex	NEUMF	Recall@5	0.7193	0.564	0.5635	0.564	-21.60%
Qwen3-complex	NEUMF	Recall@10	0.8818	0.6249	0.6261	0.6246	-29.20%
Qwen3-complex	NEUMF	Ndcg@5	0.7068	0.8557	0.8568	0.8593	+21.6%
Qwen3-complex	NEUMF	Ndcg@10	0.7892	0.845	0.8489	0.8452	+7.1%
Qwen3-complex	NEUMF	Auc	0.5053	0.7221	0.7081	0.7187	+42.2%
Qwen3-complex	NEUMF	Mrr	0.7369	0.9223	0.9302	0.9301	+26.2%
Qwen3-complex	NEUMF	HIT@RATE@5	0.8826	1	1	0.9983	+13.1%
Qwen3-complex	NEUMF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-complex	NEUMF	Map@5	0.6509	0.9045	0.9076	0.9068	+39.3%
Qwen3-complex	NEUMF	Map@10	0.6133	0.8845	0.8857	0.8863	+44.5%
Qwen3-complex	NGCF	Accuracy	0.5066	0.6874	0.6885	0.6841	+35.0%
Qwen3-complex	NGCF	Precision@5	0.4879	0.8585	0.8526	0.8549	+75.2%
Qwen3-complex	NGCF	Precision@10	0.4897	0.8367	0.8392	0.839	+71.3%
Qwen3-complex	NGCF	Recall@5	0.7139	0.5642	0.5639	0.5638	-21.00%
Qwen3-complex	NGCF	Recall@10	0.8811	0.6248	0.6256	0.6256	-29.00%
Qwen3-complex	NGCF	Ndcg@5	0.6979	0.8634	0.8564	0.8599	+23.2%
Qwen3-complex	NGCF	Ndcg@10	0.7831	0.8499	0.8491	0.8504	+8.6%
Qwen3-complex	NGCF	Auc	0.5005	0.7393	0.739	0.7355	+46.9%
Qwen3-complex	NGCF	Mrr	0.7293	0.9372	0.933	0.9377	+28.6%
Qwen3-complex	NGCF	HIT@RATE@5	0.8823	1	1	0.9983	+13.2%
Qwen3-complex	NGCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-complex	NGCF	Map@5	0.636	0.9117	0.9055	0.9079	+42.7%
Qwen3-complex	NGCF	Map@10	0.6064	0.8892	0.8846	0.8871	+46.3%
Qwen3-complex	PINSAGE	Accuracy	0.5069	0.6877	0.6833	0.6828	+34.7%
Qwen3-complex	PINSAGE	Precision@5	0.4906	0.8569	0.8569	0.8592	+75.1%
Qwen3-complex	PINSAGE	Precision@10	0.4899	0.8365	0.8413	0.839	+71.3%
Qwen3-complex	PINSAGE	Recall@5	0.7181	0.564	0.5639	0.5641	-21.40%
Qwen3-complex	PINSAGE	Recall@10	0.8816	0.6253	0.6258	0.6257	-29.00%
Qwen3-complex	PINSAGE	Ndcg@5	0.7018	0.8603	0.8591	0.8631	+23.0%
Qwen3-complex	PINSAGE	Ndcg@10	0.7848	0.8481	0.8511	0.8511	+8.4%
Qwen3-complex	PINSAGE	Auc	0.5054	0.7392	0.7323	0.7333	+45.1%

Qwen3-complex	PINSAGE	Mrr	0.7305	0.931	0.93	0.9357	+28.1%
Qwen3-complex	PINSAGE	HIT@RATE@5	0.8847	1	1	1	+13.0%
Qwen3-complex	PINSAGE	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-complex	PINSAGE	Map@5	0.6462	0.9072	0.9064	0.9094	+40.7%
Qwen3-complex	PINSAGE	Map@10	0.6161	0.8867	0.8874	0.8889	+44.3%
Qwen3-complex	SASREC	Accuracy	0.5069	0.6825	0.6811	0.6756	+33.3%
Qwen3-complex	SASREC	Precision@5	0.4907	0.8552	0.8536	0.8502	+73.3%
Qwen3-complex	SASREC	Precision@10	0.4895	0.842	0.8412	0.8393	+71.5%
Qwen3-complex	SASREC	Recall@5	0.7185	0.5639	0.5643	0.5639	-21.50%
Qwen3-complex	SASREC	Recall@10	0.8809	0.6258	0.626	0.6253	-29.00%
Qwen3-complex	SASREC	Ndcg@5	0.7048	0.8585	0.8566	0.8544	+21.2%
Qwen3-complex	SASREC	Ndcg@10	0.7872	0.8516	0.8503	0.8488	+7.8%
Qwen3-complex	SASREC	Auc	0.4981	0.7286	0.7276	0.722	+44.9%
Qwen3-complex	SASREC	Mrr	0.7351	0.9249	0.9272	0.9273	+26.1%
Qwen3-complex	SASREC	HIT@RATE@5	0.887	0.9983	1	0.9983	+12.6%
Qwen3-complex	SASREC	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-complex	SASREC	Map@5	0.6446	0.9082	0.9049	0.9049	+40.4%
Qwen3-complex	SASREC	Map@10	0.6096	0.8883	0.8854	0.8845	+45.1%
Qwen3-complex	SRGNN	Accuracy	0.5081	0.6876	0.6773	0.6789	+33.6%
Qwen3-complex	SRGNN	Precision@5	0.4896	0.8605	0.8555	0.8562	+74.9%
Qwen3-complex	SRGNN	Precision@10	0.4898	0.8374	0.841	0.8383	+71.2%
Qwen3-complex	SRGNN	Recall@5	0.716	0.5646	0.5644	0.5636	-21.30%
Qwen3-complex	SRGNN	Recall@10	0.8815	0.6251	0.6252	0.6255	-29.00%
Qwen3-complex	SRGNN	Ndcg@5	0.7006	0.8625	0.857	0.8597	+22.7%
Qwen3-complex	SRGNN	Ndcg@10	0.7845	0.8491	0.8497	0.8496	+8.3%
Qwen3-complex	SRGNN	Auc	0.5021	0.7359	0.7216	0.725	+44.4%
Qwen3-complex	SRGNN	Mrr	0.7334	0.9282	0.9255	0.9306	+26.9%
Qwen3-complex	SRGNN	HIT@RATE@5	0.8838	1	0.9983	0.9983	+13.0%
Qwen3-complex	SRGNN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-complex	SRGNN	Map@5	0.6472	0.9094	0.9029	0.907	+40.1%
Qwen3-complex	SRGNN	Map@10	0.6192	0.8885	0.8851	0.8868	+43.2%
Qwen3-structured	BPR	Accuracy	0.7098	0.6704	0.6551	0.66	-7.00%
Qwen3-structured	BPR	Precision@5	0.7826	0.843	0.8374	0.838	+7.1%
Qwen3-structured	BPR	Precision@10	0.7591	0.8349	0.8193	0.8149	+7.3%
Qwen3-structured	BPR	Recall@5	0.4386	0.5631	0.563	0.5631	+28.4%
Qwen3-structured	BPR	Recall@10	0.544	0.6252	0.6233	0.6238	+14.7%
Qwen3-structured	BPR	Ndcg@5	0.8322	0.846	0.8414	0.842	+1.2%
Qwen3-structured	BPR	Ndcg@10	0.8284	0.8428	0.8307	0.8276	-0.10%
Qwen3-structured	BPR	Auc	0.5	0.7113	0.6937	0.6937	+38.7%
Qwen3-structured	BPR	Mrr	0.5	0.9198	0.9212	0.9195	+83.9%
Qwen3-structured	BPR	HIT@RATE@5	0.9938	1	1	0.9983	+0.5%
Qwen3-structured	BPR	HIT@RATE@10	0.9987	1	1	1	+0.1%
Qwen3-structured	BPR	Map@5	0.7826	0.9003	0.8965	0.8967	+14.6%
Qwen3-structured	BPR	Map@10	0.7591	0.8791	0.8736	0.8709	+14.7%
Qwen3-structured	COMPGCN	Accuracy	0.5069	0.6888	0.6873	0.6847	+35.1%
Qwen3-structured	COMPGCN	Precision@5	0.4887	0.8562	0.8549	0.8542	+74.8%
Qwen3-structured	COMPGCN	Precision@10	0.4899	0.8382	0.8369	0.8398	+71.4%
Qwen3-structured	COMPGCN	Recall@5	0.7148	0.5642	0.5638	0.5645	-21.00%
Qwen3-structured	COMPGCN	Recall@10	0.8816	0.6255	0.6252	0.6258	-29.00%
Qwen3-structured	COMPGCN	Ndcg@5	0.6996	0.8588	0.8588	0.859	+22.8%
Qwen3-structured	COMPGCN	Ndcg@10	0.7845	0.8487	0.8483	0.8508	+8.5%
Qwen3-structured	COMPGCN	Auc	0.501	0.741	0.7387	0.734	+46.5%
Qwen3-structured	COMPGCN	Mrr	0.7302	0.9282	0.9316	0.9339	+27.9%
Qwen3-structured	COMPGCN	HIT@RATE@5	0.8815	1	0.9983	1	+13.4%
Qwen3-structured	COMPGCN	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-structured	COMPGCN	Map@5	0.6392	0.906	0.9069	0.9085	+42.1%
Qwen3-structured	COMPGCN	Map@10	0.6077	0.8868	0.8869	0.8871	+46.0%
Qwen3-structured	DEEPPM	Accuracy	0.5055	0.6786	0.6749	0.6858	+35.7%
Qwen3-structured	DEEPPM	Precision@5	0.4898	0.8529	0.8499	0.8555	+74.7%

Qwen3-structured	DEEPPFM	Precision@10	0.4899	0.8372	0.8387	0.8362	+70.7%
Qwen3-structured	DEEPPFM	Recall@5	0.7158	0.564	0.5639	0.564	-21.20%
Qwen3-structured	DEEPPFM	Recall@10	0.8815	0.6254	0.6255	0.6253	-29.10%
Qwen3-structured	DEEPPFM	Ndcg@5	0.7007	0.8609	0.8546	0.8601	+22.8%
Qwen3-structured	DEEPPFM	Ndcg@10	0.7847	0.8507	0.8486	0.8485	+8.1%
Qwen3-structured	DEEPPFM	Auc	0.5047	0.7279	0.7191	0.7332	+45.3%
Qwen3-structured	DEEPPFM	Mrr	0.7297	0.9368	0.9295	0.9339	+28.0%
Qwen3-structured	DEEPPFM	HIT@RATE@5	0.8828	0.9983	1	1	+13.3%
Qwen3-structured	DEEPPFM	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-structured	DEEPPFM	Map@5	0.643	0.9141	0.9071	0.9086	+41.3%
Qwen3-structured	DEEPPFM	Map@10	0.6075	0.8895	0.8854	0.8877	+46.1%
Qwen3-structured	GAT	Accuracy	0.5069	0.688	0.6788	0.679	+33.9%
Qwen3-structured	GAT	Precision@5	0.4881	0.8582	0.8526	0.8582	+75.8%
Qwen3-structured	GAT	Precision@10	0.4896	0.8402	0.8379	0.8369	+70.9%
Qwen3-structured	GAT	Recall@5	0.715	0.5643	0.5642	0.5644	-21.10%
Qwen3-structured	GAT	Recall@10	0.8811	0.6259	0.6251	0.6252	-29.00%
Qwen3-structured	GAT	Ndcg@5	0.698	0.8599	0.8547	0.8608	+23.3%
Qwen3-structured	GAT	Ndcg@10	0.7827	0.8502	0.847	0.8486	+8.4%
Qwen3-structured	GAT	Auc	0.4989	0.7388	0.7263	0.7302	+46.4%
Qwen3-structured	GAT	Mrr	0.7279	0.9292	0.9228	0.9303	+27.8%
Qwen3-structured	GAT	HIT@RATE@5	0.884	1	1	1	+13.1%
Qwen3-structured	GAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	GAT	Map@5	0.643	0.9065	0.9038	0.9068	+41.0%
Qwen3-structured	GAT	Map@10	0.6052	0.8867	0.8835	0.8886	+46.8%
Qwen3-structured	GCN	Accuracy	0.5069	0.6836	0.6794	0.677	+33.6%
Qwen3-structured	GCN	Precision@5	0.4901	0.8552	0.8572	0.8536	+74.2%
Qwen3-structured	GCN	Precision@10	0.4899	0.8383	0.8354	0.8365	+70.8%
Qwen3-structured	GCN	Recall@5	0.7181	0.5639	0.5642	0.5636	-21.50%
Qwen3-structured	GCN	Recall@10	0.8817	0.6253	0.6255	0.6252	-29.10%
Qwen3-structured	GCN	Ndcg@5	0.7019	0.8589	0.8618	0.8586	+22.3%
Qwen3-structured	GCN	Ndcg@10	0.7852	0.8492	0.8488	0.8485	+8.1%
Qwen3-structured	GCN	Auc	0.5027	0.7325	0.7242	0.7191	+43.0%
Qwen3-structured	GCN	Mrr	0.7307	0.9267	0.9341	0.9342	+27.8%
Qwen3-structured	GCN	HIT@RATE@5	0.8855	0.9983	1	1	+12.9%
Qwen3-structured	GCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	GCN	Map@5	0.6434	0.9075	0.9103	0.9083	+41.2%
Qwen3-structured	GCN	Map@10	0.6171	0.8861	0.889	0.8864	+43.6%
Qwen3-structured	GRAPHSAGE	Accuracy	0.6908	0.6836	0.6748	0.6752	-2.30%
Qwen3-structured	GRAPHSAGE	Precision@5	0.7639	0.8529	0.8552	0.8565	+12.1%
Qwen3-structured	GRAPHSAGE	Precision@10	0.7393	0.8387	0.8365	0.8388	+13.5%
Qwen3-structured	GRAPHSAGE	Recall@5	0.4333	0.5634	0.5639	0.5637	+30.1%
Qwen3-structured	GRAPHSAGE	Recall@10	0.5326	0.6257	0.6254	0.6252	+17.4%
Qwen3-structured	GRAPHSAGE	Ndcg@5	0.813	0.8586	0.8604	0.8575	+5.5%
Qwen3-structured	GRAPHSAGE	Ndcg@10	0.8052	0.8503	0.8491	0.8483	+5.4%
Qwen3-structured	GRAPHSAGE	Auc	0.5	0.7285	0.7251	0.7186	+43.7%
Qwen3-structured	GRAPHSAGE	Mrr	0.5	0.9337	0.9361	0.9267	+85.3%
Qwen3-structured	GRAPHSAGE	HIT@RATE@5	0.9931	0.9983	1	1	+0.7%
Qwen3-structured	GRAPHSAGE	HIT@RATE@10	0.9976	1	1	1	+0.2%
Qwen3-structured	GRAPHSAGE	Map@5	0.7639	0.9093	0.909	0.903	+18.2%
Qwen3-structured	GRAPHSAGE	Map@10	0.7393	0.8874	0.8878	0.886	+19.8%
Qwen3-structured	LIGHTGCN	Accuracy	0.5076	0.6806	0.6738	0.6819	+34.3%
Qwen3-structured	LIGHTGCN	Precision@5	0.491	0.8516	0.8463	0.8545	+74.0%
Qwen3-structured	LIGHTGCN	Precision@10	0.4896	0.8387	0.8312	0.8367	+70.9%
Qwen3-structured	LIGHTGCN	Recall@5	0.7187	0.5636	0.5629	0.5638	-21.50%
Qwen3-structured	LIGHTGCN	Recall@10	0.8809	0.6254	0.624	0.6253	-29.00%
Qwen3-structured	LIGHTGCN	Ndcg@5	0.7051	0.8562	0.8518	0.8556	+21.3%
Qwen3-structured	LIGHTGCN	Ndcg@10	0.7871	0.849	0.8429	0.8462	+7.5%
Qwen3-structured	LIGHTGCN	Auc	0.5013	0.73	0.7142	0.7291	+45.4%
Qwen3-structured	LIGHTGCN	Mrr	0.7348	0.9308	0.9304	0.9217	+25.4%

Qwen3-structured	LIGHTGCN	HIT@RATE@5	0.8855	1	1	1	+12.9%
Qwen3-structured	LIGHTGCN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	LIGHTGCN	Map@5	0.647	0.9075	0.9056	0.9024	+39.5%
Qwen3-structured	LIGHTGCN	Map@10	0.6108	0.8854	0.8824	0.8835	+44.6%
Qwen3-structured	KGAT	Accuracy	0.5069	0.6881	0.6814	0.6786	+33.9%
Qwen3-structured	KGAT	Precision@5	0.4881	0.8612	0.8555	0.8552	+75.2%
Qwen3-structured	KGAT	Precision@10	0.4896	0.8393	0.8355	0.8359	+70.7%
Qwen3-structured	KGAT	Recall@5	0.715	0.5642	0.5642	0.5638	-21.20%
Qwen3-structured	KGAT	Recall@10	0.8811	0.6254	0.6249	0.6253	-29.00%
Qwen3-structured	KGAT	Ndcg@5	0.698	0.8644	0.8592	0.857	+22.8%
Qwen3-structured	KGAT	Ndcg@10	0.7827	0.8515	0.8474	0.8465	+8.1%
Qwen3-structured	KGAT	Auc	0.4989	0.7395	0.7293	0.7253	+45.4%
Qwen3-structured	KGAT	Mrr	0.7279	0.9315	0.9278	0.9241	+27.0%
Qwen3-structured	KGAT	HIT@RATE@5	0.884	0.9983	1	1	+13.1%
Qwen3-structured	KGAT	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	KGAT	Map@5	0.643	0.9108	0.9083	0.9052	+40.8%
Qwen3-structured	KGAT	Map@10	0.6052	0.8899	0.8867	0.8852	+46.3%
Qwen3-structured	KGIN	Accuracy	0.5069	0.6898	0.6852	0.6842	+35.0%
Qwen3-structured	KGIN	Precision@5	0.4906	0.8605	0.8562	0.8595	+75.2%
Qwen3-structured	KGIN	Precision@10	0.4899	0.8347	0.8374	0.8354	+70.5%
Qwen3-structured	KGIN	Recall@5	0.7181	0.5641	0.564	0.5647	-21.40%
Qwen3-structured	KGIN	Recall@10	0.8816	0.6243	0.6256	0.625	-29.10%
Qwen3-structured	KGIN	Ndcg@5	0.7018	0.8627	0.8588	0.8627	+22.9%
Qwen3-structured	KGIN	Ndcg@10	0.7848	0.8473	0.8482	0.8483	+8.1%
Qwen3-structured	KGIN	Auc	0.5054	0.7401	0.7358	0.7343	+45.3%
Qwen3-structured	KGIN	Mrr	0.7305	0.9291	0.9282	0.935	+28.0%
Qwen3-structured	KGIN	HIT@RATE@5	0.8847	1	1	1	+13.0%
Qwen3-structured	KGIN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	KGIN	Map@5	0.6462	0.9094	0.9064	0.9108	+40.9%
Qwen3-structured	KGIN	Map@10	0.6161	0.8877	0.8857	0.889	+44.3%
Qwen3-structured	MULTIGCCF	Accuracy	0.5066	0.6895	0.6857	0.6862	+35.4%
Qwen3-structured	MULTIGCCF	Precision@5	0.4879	0.8565	0.8549	0.8592	+76.1%
Qwen3-structured	MULTIGCCF	Precision@10	0.4897	0.8332	0.839	0.8359	+70.7%
Qwen3-structured	MULTIGCCF	Recall@5	0.7139	0.5638	0.564	0.5648	-20.90%
Qwen3-structured	MULTIGCCF	Recall@10	0.8811	0.6251	0.6254	0.6251	-29.10%
Qwen3-structured	MULTIGCCF	Ndcg@5	0.6979	0.8569	0.8608	0.8633	+23.7%
Qwen3-structured	MULTIGCCF	Ndcg@10	0.7831	0.844	0.8508	0.849	+8.4%
Qwen3-structured	MULTIGCCF	Auc	0.5005	0.7406	0.736	0.7344	+46.7%
Qwen3-structured	MULTIGCCF	Mrr	0.7293	0.9224	0.932	0.9366	+28.4%
Qwen3-structured	MULTIGCCF	HIT@RATE@5	0.8823	0.9983	0.9983	0.9983	+13.2%
Qwen3-structured	MULTIGCCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-structured	MULTIGCCF	Map@5	0.636	0.9021	0.914	0.9109	+43.2%
Qwen3-structured	MULTIGCCF	Map@10	0.6064	0.8837	0.8889	0.8891	+46.6%
Qwen3-structured	NEUMF	Accuracy	0.5069	0.6834	0.6815	0.6784	+33.8%
Qwen3-structured	NEUMF	Precision@5	0.4917	0.8529	0.8516	0.8569	+74.3%
Qwen3-structured	NEUMF	Precision@10	0.4901	0.8352	0.8339	0.8347	+70.3%
Qwen3-structured	NEUMF	Recall@5	0.7193	0.5638	0.5636	0.5644	-21.50%
Qwen3-structured	NEUMF	Recall@10	0.8818	0.6248	0.6246	0.6252	-29.10%
Qwen3-structured	NEUMF	Ndcg@5	0.7068	0.8571	0.8538	0.862	+22.0%
Qwen3-structured	NEUMF	Ndcg@10	0.7892	0.8467	0.8442	0.8481	+7.5%
Qwen3-structured	NEUMF	Auc	0.5053	0.7291	0.7266	0.7261	+43.7%
Qwen3-structured	NEUMF	Mrr	0.7369	0.9324	0.921	0.935	+26.9%
Qwen3-structured	NEUMF	HIT@RATE@5	0.8826	1	0.9983	0.9967	+12.9%
Qwen3-structured	NEUMF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-structured	NEUMF	Map@5	0.6509	0.9076	0.9033	0.9098	+39.8%
Qwen3-structured	NEUMF	Map@10	0.6133	0.8858	0.8842	0.8868	+44.6%
Qwen3-structured	NGCF	Accuracy	0.5066	0.6884	0.6912	0.687	+35.6%
Qwen3-structured	NGCF	Precision@5	0.4879	0.8549	0.8552	0.8588	+76.0%
Qwen3-structured	NGCF	Precision@10	0.4897	0.836	0.8395	0.8359	+70.7%

Qwen3-structured	NGCF	Recall@5	0.7139	0.5638	0.5642	0.5643	-21.00%
Qwen3-structured	NGCF	Recall@10	0.8811	0.625	0.626	0.6252	-29.00%
Qwen3-structured	NGCF	Ndcg@5	0.6979	0.8579	0.8576	0.8601	+23.2%
Qwen3-structured	NGCF	Ndcg@10	0.7831	0.8471	0.8494	0.8471	+8.2%
Qwen3-structured	NGCF	Auc	0.5005	0.7397	0.7428	0.7363	+47.1%
Qwen3-structured	NGCF	Mrr	0.7293	0.9266	0.928	0.9277	+27.2%
Qwen3-structured	NGCF	HIT@RATE@5	0.8823	1	0.9983	1	+13.3%
Qwen3-structured	NGCF	HIT@RATE@10	0.8952	1	1	1	+11.7%
Qwen3-structured	NGCF	Map@5	0.636	0.9061	0.9059	0.9071	+42.6%
Qwen3-structured	NGCF	Map@10	0.6064	0.8853	0.8863	0.8866	+46.2%
Qwen3-structured	PINSAGE	Accuracy	0.5069	0.6867	0.6869	0.6886	+35.9%
Qwen3-structured	PINSAGE	Precision@5	0.4906	0.8549	0.8621	0.8565	+74.6%
Qwen3-structured	PINSAGE	Precision@10	0.4899	0.8383	0.8337	0.8383	+71.1%
Qwen3-structured	PINSAGE	Recall@5	0.7181	0.5641	0.5647	0.5644	-21.40%
Qwen3-structured	PINSAGE	Recall@10	0.8816	0.6257	0.6245	0.6257	-29.00%
Qwen3-structured	PINSAGE	Ndcg@5	0.7018	0.8589	0.8634	0.859	+22.4%
Qwen3-structured	PINSAGE	Ndcg@10	0.7848	0.8494	0.8465	0.8488	+8.2%
Qwen3-structured	PINSAGE	Auc	0.5054	0.7385	0.7328	0.7388	+46.2%
Qwen3-structured	PINSAGE	Mrr	0.7305	0.9287	0.9311	0.9284	+27.1%
Qwen3-structured	PINSAGE	HIT@RATE@5	0.8847	1	1	1	+13.0%
Qwen3-structured	PINSAGE	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	PINSAGE	Map@5	0.6462	0.9098	0.9069	0.9067	+40.3%
Qwen3-structured	PINSAGE	Map@10	0.6161	0.8874	0.8879	0.886	+43.8%
Qwen3-structured	SASREC	Accuracy	0.5069	0.6831	0.6767	0.6751	+33.2%
Qwen3-structured	SASREC	Precision@5	0.4907	0.8539	0.8542	0.8555	+74.4%
Qwen3-structured	SASREC	Precision@10	0.4895	0.839	0.8426	0.8423	+72.1%
Qwen3-structured	SASREC	Recall@5	0.7185	0.5644	0.5637	0.5642	-21.50%
Qwen3-structured	SASREC	Recall@10	0.8809	0.6256	0.6257	0.6259	-29.00%
Qwen3-structured	SASREC	Ndcg@5	0.7048	0.8545	0.8577	0.8545	+21.2%
Qwen3-structured	SASREC	Ndcg@10	0.7872	0.8471	0.8518	0.8491	+7.9%
Qwen3-structured	SASREC	Auc	0.4981	0.73	0.7241	0.7236	+45.3%
Qwen3-structured	SASREC	Mrr	0.7351	0.9191	0.9294	0.9164	+24.7%
Qwen3-structured	SASREC	HIT@RATE@5	0.887	1	1	0.9983	+12.6%
Qwen3-structured	SASREC	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	SASREC	Map@5	0.6446	0.9019	0.9067	0.9004	+39.7%
Qwen3-structured	SASREC	Map@10	0.6096	0.8833	0.8875	0.8846	+45.1%
Qwen3-structured	SRGNN	Accuracy	0.5081	0.6863	0.6823	0.6833	+34.5%
Qwen3-structured	SRGNN	Precision@5	0.4896	0.8598	0.8512	0.8575	+75.1%
Qwen3-structured	SRGNN	Precision@10	0.4898	0.8403	0.836	0.837	+70.9%
Qwen3-structured	SRGNN	Recall@5	0.716	0.5643	0.5637	0.5643	-21.20%
Qwen3-structured	SRGNN	Recall@10	0.8815	0.626	0.6252	0.6258	-29.00%
Qwen3-structured	SRGNN	Ndcg@5	0.7006	0.8608	0.8583	0.8628	+23.2%
Qwen3-structured	SRGNN	Ndcg@10	0.7845	0.8501	0.8489	0.8502	+8.4%
Qwen3-structured	SRGNN	Auc	0.5021	0.7386	0.7292	0.7283	+45.1%
Qwen3-structured	SRGNN	Mrr	0.7334	0.9277	0.9354	0.9381	+27.9%
Qwen3-structured	SRGNN	HIT@RATE@5	0.8838	0.9983	0.9983	0.9983	+13.0%
Qwen3-structured	SRGNN	HIT@RATE@10	0.8954	1	1	1	+11.7%
Qwen3-structured	SRGNN	Map@5	0.6472	0.9049	0.9118	0.9098	+40.6%
Qwen3-structured	SRGNN	Map@10	0.6192	0.887	0.8879	0.8887	+43.5%